# Fifteen Code-Based KEMs, One Framework: Structural and Parametric Cryptanalysis of the NGCC Round-1 Code-Based Candidates*

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**Abstract**

The Chinese Next-Generation Commercial Cryptography (NGCC) competition received fifteen code-based key encapsulation mechanisms and one code-based signature in its first round (September 2026). We give the first unified design-level analysis of all of them: eight quasi-cyclic Hamming-metric KEMs (HARE, HEP-QC, NSS-HQC, TriQ-KEM, QUBE, Mito, BIKE-MLThre, TRIKE), one quasi-cyclic alternant/Goppa KEM (QCTM), five rank-metric KEMs (BRQC, BRA, C-Multi-UR-AG, BAG-Loong, BAG-Piglet) and the UVW KEM and signature. Every candidate is re-priced with one methodology (CryptographicEstimators 2.1.1 for information-set and rank decoding, the May–Sá Diogo same-key multi-instance DOOM estimator, and HQC's independent-coordinate failure model), and every published forum finding is recorded, with the parametric ones independently recomputed.

Our main structural result concerns QCTM. We prove that its "twisted" parity row is a combination of the ordinary rows, so that the public code is a quasi-cyclic *alternant* code correcting only half the errors of a Goppa code of the same redundancy; that its fold is *exactly* the binary Goppa code $\Gamma(X', M) \cap E$ with support in the subgroup of 19-th powers (as observed empirically on the forum); and, new, that any quotient key—in any of the equivalent forms a Goppa key-recovery algorithm may return—*lifts in polynomial time* to a decoding key for QCTM, by a character-projection linear system that we verify on scaled instances. QCTM's key security is therefore polynomially equivalent to Goppa key recovery at $(m, n, t) = (18, n/19, t/19)$, which closes the gap left open on the tracker and makes the published heuristic quotient estimates ($2^{141}$–$2^{142}$ for QCTM256/512) estimates for QCTM itself.

On the parametric side we find that TriQ-KEM's own natural decryption-failure rates (its specification's Table-6 values $2^{-121.9}, 2^{-173.4}, 2^{-209.6}, 2^{-230.0}$) fall short of a generic $q/2^{\lambda}$ proof term at every level, though they meet the specification's own stated criterion; that TRIKE-1's message security drops below $2^{128}$ after only $2^{31}$ ciphertexts under a static key, with TRIKE-3/5/7/9, TriQ-KEM, QUBE, HARE-1 and NSS-HQC-128 crossing their targets within the $2^{80}$ budget; that BIKE-MLThre-512 abandons the "2 primitive mod $r$" design rule; and that, in the regime where the blockwise MaxMinors bound $\overline{E}$ is tight (the designers' block order, away from heavy puncturing, per Song et al.), an independent re-implementation of the BRQC and BRA cost reproduces the designers' published strengths and finds no point below them—correcting an earlier draft that had reported sub-tabulated points from evaluating $\overline{E}$ outside its tight

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regime. All rank-metric sets clear their levels by 53 bits or more. For the already-broken BAG-Loong we additionally execute, on its reference implementation, the implicit-rejection oracle the tracker had downgraded to a design concern, confirming the KEM is not IND-CCA2 independently of the sampler defect that already recovers its key. The outcome is a four-way classification (previously broken / newly broken or reduced / below the claimed level / no design-level weakness found) with every entry labelled as proved, measured, computed or cited.
**Keywords:** code-based cryptography · NGCC · quasi-cyclic codes · twisted Goppa codes · folding · rank metric · information-set decoding · decryption failure

# 1 Introduction

The Institute of Commercial Cryptography Standards (ICCS) published the Round-1 candidates of its Next-Generation Commercial Cryptography (NGCC) programme on 20 September 2026: 41 KEMs, 34 signatures and 9 key-exchange protocols. Fifteen of the KEMs and one signature are code-based. Unlike the NIST process, the NGCC call provided a short public window and a public forum, and within six weeks an unofficial tracker (`ngcc.dev`, maintained by M.-J. O. Saarinen) had catalogued 430 findings across 111 of the 119 candidates. Of the 365 active public-key findings, 187 were implementation defects (many from AI-assisted sweeps) and 149 were design-level, 69 of those rated critical and 57 confirmed—so design-level results are a substantial fraction, not a handful. The code-based candidates are, with one exception, fresh designs with no prior third-party cryptanalysis: eight quasi-cyclic Hamming-metric KEMs in the HQC or BIKE lineage (HARE, HEP-QC, NSS-HQC, TriQ-KEM, QUBE, Mito, BIKE-MLThre, TRIKE), one Niederreiter-type KEM built on quasi-cyclic *twisted* Goppa codes (QCTM), five rank-metric KEMs with blockwise or multi-syndrome errors (BRQC, BRA, C-Multi-UR-AG, BAG-Loong, BAG-Piglet), and the UVW KEM, whose structure (masked Reed–Solomon codes in a $(U{+}V, U{+}W)$ construction) was broken in our companion work and on the forum.

**What this paper does.** We treat the fifteen KEMs (and the UVW signature) with one methodology and report, for every parameter set, (a) what the forum had established by 31 October 2026, with attribution, (b) what we add, and (c) a verdict in one of four classes: *previously broken*, *newly broken or reduced here*, *not as secure as claimed*, and *no design-level weakness found*. Every quantitative claim carries one of four labels. [Proved] means a statement with a proof in this paper. [Measured] means an experiment we ran (the script for each is named in Appendix C). [Computed] means a number produced by a public estimator at pinned versions—CryptographicEstimators 2.1.1 [7] for syndrome decoding (Prange, Stern, Dumer, ball-collision, BJMM, May–Ozerov) and rank syndrome decoding (GRS/OJ enumeration, MaxMinors, SupportMinors), and the DS-DOOM estimator of May and Sá Diogo [8] (commit `39b78dc`, `doom.py` SHA-256 `d53e8bb0...`) for the same-key multi-ciphertext setting; the HQC Model-1 failure calculation is our own implementation validated against the HQC and QUBE specifications to the last printed digit. [Cited] means a forum or ePrint result we did not reproduce.

**Contributions.**

- **QCTM is a quasi-cyclic alternant code whose fold lifts** (§4). We prove that the twisted row of QCTM's parity-check matrix equals $\eta\mathbf{1}$ plus a combination of the ordinary rows (Lemma 3), so the public code is the alternant code $\mathcal{A}_{t-1}(L, g^{-1}) \cap E$ (Theorem 4); that its fold is exactly the Goppa code $\Gamma(X', M) \cap E$ with $X' = \{a_O^{\ell}\}$ (Theorem 5), which gives a proof of the empirical observation kem-32-4; and that a quotient key lifts in polynomial time to a decoding key for QCTM

(Theorem 7), the step the tracker lists as missing. The lifting uses a character projection of the public dual and a linear system in the orbit rotations; it is verified end-to-end on scaled instances. QCTM's key security is thus *equivalent* to Goppa key recovery at $(18, 530, 15)$, $(18, 1000, 27)$, $(18, 2000, 53)$ with subgroup support, and the published heuristic quotient estimates become estimates for QCTM.

- **Uniform re-pricing of the quasi-cyclic family** (§3). A ring audit ($x^n - 1$ factorisation), single-instance ISD costs for key and message recovery, DS-DOOM multi-instance crossings, and HQC Model-1 failure rates for every parameter set of the eight QC KEMs, side by side with HQC. New same-key crossings (filed on the tracker as kem-26-6, kem-33-5, kem-36-9, kem-37-3, and the NSS-HQC static-key estimate kem-26-7): TRIKE's own (unsubmitted) level-1 set crosses $2^{128}$ after $2^{31}$ ciphertexts; TriQ-KEM, QUBE and NSS-HQC-128 cross within $2^{80}$. These are model crossings under the same estimator-and-key-mapping standard as the earlier BIKE-MLThre/HARE/QCTM findings, not completed full-size runs; several cross by well under a bit and assume many independently generated ciphertexts under one static key whose errors the attacker does not know. TriQ-KEM's natural failure rates (its specification's Table-6 values) fall 6 to 282 bits short of a generic $q/2^\lambda$ proof term, though they meet the specification's own stated criterion; this is a proof-term observation, not a demonstrated failure attack. BIKE-MLThre-512 uses $r = 150001$ with $\mathrm{ord}_r(2) = 3750$, so $x^r - 1$ has 40 non-trivial factors, against the BIKE design rule (we found no attack, and file none). Mito's 2-adic structure costs it about two bits and leaves it with large margins; HEP-QC's disguised code is broken (kem-17-4) but its KEM security is exactly HQC's.
- **Rank metric** (§5). We re-implement the blockwise MaxMinors cost of BRQC/BRA (Theorems 7.6/7.7 of the specifications) and validate it against the designers' own data points. Restricted to the regime in which $\overline{E}$ is a tight bound—the designers' largest-rank-first block order, away from degenerate heavy puncturing, as established by Song et al. [26]—our re-implementation reproduces the published strengths 181/295/566 (BRQC) and 189/304/556 (BRA) and finds no point below them; an earlier free-ordering, heavy-puncturing search that appeared to beat the tables evaluated $\overline{E}$ outside its tight regime and is withdrawn. We re-price C-Multi-UR-AG with CryptographicEstimators and show by experiment that stacking MaxMinors equations of the $N$ same-support syndromes saturates at $m\binom{n-k}{r} - m$, so the multi-syndrome structure gives at most $(n-k)/(n-k-r)$; we confirm the merged-support shortfall of BAG-Loong (kem-03-4) with slightly lower numbers. We also *execute*, on BAG-Loong's own reference implementation, the implicit-rejection oracle the tracker had filed against it only as a Medium design concern (kem-03-2): 4000 distinct corrupted ciphertexts return one shared secret, which depends only on the decrypted message, so the KEM is not IND-CCA2 at the design level ([Measured]); BAG-Piglet was already broken the same way (kem-04-2).
- **A complete, attributed status table** (Table 2) and a list of what remains open.

**What we do not claim.** Our one executed attack is the BAG-Loong implicit-rejection oracle (§5.4.1); it reproduces, on that candidate's reference code, a flaw the tracker had already confirmed-and-executed on the sibling BAG-Piglet, and we do not carry it through to a full secret-key read-out (its key is already recoverable by the unrelated defect kem-03-3). We did not run any attack at full size against a scheme the forum classifies as structurally unbroken. Our other new results are a reduction (QCTM) and parameter shortfalls under pinned public estimators (multi-instance, failure rates, the NSS-HQC key extrapolation). Where a forum finding is a *lead* rather than a confirmed result we say so, and where a heuristic cost model is involved (the 2026 Goppa key-recovery models, the DS-DOOM model) we report the model and its margins rather than a verdict.

Table 1: Triage: current status of the fifteen KEMs. The first group is broken—a key or session key is recovered, or the IND-CCA2 claim fails at the design level—and no choice of parameters repairs it. The second group meets no attack structurally but misses its claimed bit-security at some levels; a heavier parameter set (or, for the multi-ciphertext crossings, a per-key session cap) would plausibly restore the claim. The third group withstands our analysis at its submitted parameters.

| Status | Candidates | Outlook |
|---|---|---|
| **Broken** | BAG-Piglet, BAG-Loong, UVW-KEM, UVW-signature, HEP-QC, NSS-HQC | not repairable by re-parametrisation; exclude from further evaluation |
| **Below claimed level** | QCTM, TRIKE, QUBE, HARE, TriQ-KEM, BIKE-MLThre | may survive with larger parameters / a session cap; QCTM additionally hinges on Goppa key-recovery hardness (§4) |
| **Standing** | BRQC, BRA, C-Multi-UR-AG, Mito | no design-level weakness found; parameters already clear the claimed levels |

**Sources.** Specifications and implementations are the official Round-1 packages (mirror `Tongsuo-Project/NGCC-candidates`, catalogue of 21 October 2026); forum findings are quoted from the tracker pages `ngcc.dev/reports/kem-XX.html` as of 31 October 2026 and from the verification packages of Xiong and Wang [4] and of the tracker harness [3]. Finding identifiers such as kem-32-4 refer to the tracker.

## 2 Preliminaries and methodology

**Codes.** $\mathbb{F}_q$ is the finite field with $q$ elements; wt is the Hamming weight. For $\mathbf{x} \in \mathbb{F}_{q^m}^n$ with distinct coordinates and $\mathbf{y} \in (\mathbb{F}_{q^m}^*)^n$, $\mathrm{GRS}_r(\mathbf{x},\mathbf{y}) = \{(y_j f(x_j))_j : \deg f < r\}$, the alternant code $\mathcal{A}_r(\mathbf{x},\mathbf{y}) = \mathrm{GRS}_r(\mathbf{x},\mathbf{y})^\perp \cap \mathbb{F}_q^n$, and for $g \in \mathbb{F}_{q^m}[x]$ with $g(x_j) \neq 0$ the Goppa code is $\Gamma(\mathbf{x}, g) = \mathcal{A}_{\deg g}(\mathbf{x}, g^{-1})$. For binary Goppa codes with square-free $g$, $\Gamma(\mathbf{x}, g) = \Gamma(\mathbf{x}, g^2)$, so $\deg g$ errors are correctable. $E$ denotes the even-weight code. The trace code of $D \subseteq \mathbb{F}_{q^m}^n$ is $\mathrm{Tr}(D)$ and Delsarte's theorem gives $(D^\perp \cap \mathbb{F}_q^n)^\perp = \mathrm{Tr}(D)$. The Schur product of two codes is spanned by coordinate-wise products.

**Quasi-cyclic decoding problems.** $\mathcal{R}_n = \mathbb{F}_2[x]/(x^n-1)$. The 2-QCSD (or 2-QCCF) instance behind an HQC-type public key $(h, s = x+hy)$ is: find $(x, y)$ of weight $(w, w)$ with $[\,I \mid \mathrm{rot}(h)\,](x,y)^\top = s$, a syndrome-decoding instance in a $[2n, n]$ code of weight $2w$ with $n$ free cyclic rotations. A ciphertext $u = r_1 + hr_2$ gives the same instance with weight $2w_r$; the second ciphertext component only adds equations. The multi-block generalisations (QUBE's $(s,t)$-QCSD, TRIKE's $(2,3)$-QCSD and 3-QCCF, Mito's 2-adic rings) are listed with each scheme. We price every instance with CryptographicEstimators [7] (bit operations, memory in bits, no memory-access penalty) over Prange, Stern, Dumer, ball-collision, BJMM, BJMM$^+$ and May–Ozerov, and report the Stern-family minimum (the first four) and the overall minimum. We exclude the depth-tuned variants BJMM-dw, BJMM-pdw and Both–May, which dominate the estimator's run time and never give the minimum in the designers' own tables (e.g. QCTM's Tables 1–3, where the minima are May–Ozerov, May–Ozerov and BJMM). The quasi-cyclic discount is the designers' usual convention, $\log_2 n$ for codeword finding and $\frac{1}{2}\log_2 n$ for syndrome decoding (DOOM [9]). The configuration is calibrated against two published candidate tables. For QCTM128 ($[10070, 4957]$, $w = 142$, $\frac{1}{2}\log_2 19 = 2.12$)

we obtain Stern 154.64 → 152.52 and overall minimum May–Ozerov 152.99 → 150.87, matching every digit of that specification's Table 1 including its stated final strength of 150.87 bits, and QCTM256 comes out at 272.91 against its 272.90. For TRIKE's key instances ($\log_2 r = 13.33$ and 13.93) we obtain Stern 147.38 → 134.05 and 185.71 → 171.78, matching its Table 9 in both the raw column and the stated complexity. Our instance definitions and discount conventions are therefore the designers' own, which is what licenses us to quote their numbers for the instances we did not recompute.

**Same-key multi-instance decoding.** May and Sá Diogo [8] price the attack that collects $Q$ ciphertexts under one static public key, uses all $nQ$ rotated syndromes as targets of one decoding-one-out-of-many search and recovers the error (hence the session key) of *one* of them. Their DS-DOOM estimator, pinned at the commit used by the tracker, is applied to every candidate: for each parameter set we report the single-instance DS-DOOM cost ($M = n$ targets), and the smallest session exponent $s \leq 80$ for which $M = n2^s$ targets drive the cost below the claimed level. The NGCC call imposes no per-key session cap on KEMs; the tracker treats $2^{80}$ as the evaluation budget, and so do we. These numbers are a heuristic cost model with sub-bit margins in several cases; we never turn a crossing by a fraction of a bit into a "break".

**Failure rates.** For the RM–RS concatenated schemes we implemented HQC's Model 1 [10]: the coordinates of the decryption noise are independent Bernoulli with parameter obtained from the ring-product distribution (Proposition 6.1.1 of [10]) and the odd-sum rule; the duplicated $[n_2, 8, n_2/2]$ Reed–Muller inner code fails with the improved bound of Proposition 6.1.4 (which we transcribed from the QUBE specification where it is reproduced in full), and the outer $[n_1, k_1]$ Reed–Solomon code fails when more than $\lfloor (n_1 - k_1)/2 \rfloor$ inner blocks fail. The implementation reproduces HQC-1/3/5 ($2^{-132.892}, 2^{-193.861}, 2^{-260.597}$) and QUBE's Table 7 to the printed precision (Table 5). Model 1 ignores the dependence between coordinates; the refined models of [11] give lower failure rates, which is why several candidates quote Model 2 figures.

**Rank metric.** For $\mathbb{F}_{q^m}$-linear codes and errors of rank $r$ we use CryptographicEstimators' `RankSDEstimator` (combinatorial GRS/OJ variants and the algebraic MaxMinors/SupportMinors modellings with hybrid guessing and puncturing [22, 23]) with the linear-algebra exponent $\omega = 2.81$ unless stated; the blockwise designers use $\omega = 2$, which we reproduce separately in §5.

**Folding.** If a code $\mathcal{C}$ of length $n = \ell n_b$ is invariant under a permutation $\sigma$ whose orbits are the $n_b$ blocks of size $\ell$, the folded code is $\varphi(\mathcal{C}) = \{(\sum_{i \in O} c_i)_O : c \in \mathcal{C}\}$. For $\ell$ odd and characteristic 2 it is the compression of the $\sigma$-invariant subcode. Faugère et al. [12] proved that the fold of a quasi-cyclic alternant code (with the group acting as an affine map on the support) is again alternant; §4 makes this exact for QCTM.

## 3 The quasi-cyclic Hamming-metric family

Eight candidates are HQC- or BIKE-type KEMs over $\mathcal{R}_n = \mathbb{F}_2[x]/(x^n - 1)$. Their public keys are sparse-ring relations, their ciphertexts expose a quasi-cyclic syndrome, and their correctness rests on a concatenated or MDPC decoder with a claimed failure rate. We audit the three ingredients uniformly: the ring (§3.1), the decoding costs in the single- and multi-ciphertext settings (§3.2, §3.3), and the failure rates (§3.4); the per-scheme verdicts are in §3.5.

### 3.1 Ring audit

BIKE's design rule requires $n$ prime with 2 primitive modulo $n$, so that $x^n - 1 = (x+1)\Phi(x)$ with $\Phi$ irreducible: the ring then has no proper quotients beyond $\mathbb{F}_2$, no squaring/folding sub-structure, and a random sparse element is invertible with overwhelming probability. All 41 ring lengths of the eight QC candidates (counting Mito's three variants separately) and HQC's three are prime. All have 2 primitive except one: BIKE-MLThre-512 uses $n = 150001$ with $\mathrm{ord}_n(2) = 3750$, so that

$$x^{150001} - 1 = (x+1)\prod_{i=1}^{40} f_i(x), \qquad \deg f_i = 3750.$$

[Computed] (`ring_audit.py`, sympy). The ring therefore has $2^{41}$ idempotents and 40 field quotients $\mathbb{F}_{2^{3750}}$. We found no attack: every non-trivial quotient is a field of 3750 bits in which sparsity is not preserved, the probability that a weight-273 secret block is a zero divisor is $40 \cdot 2^{-3750}$, and the squaring attacks of Löndahl et al. need $n$ composite. But the specification neither states nor justifies the deviation, which removes the standard argument that the only algebraic structure available to an attacker is the cyclic shift; we classify it as a design-rule violation without a known exploit.

### 3.2 Single-instance decoding costs

Table 3 re-prices, with one estimator and one set of conventions, the instances for which a specification publishes a comparable number (these double as calibration) together with the lowest-level instance of each family, where the margins are thinnest, up to a size cap; for the instances above the cap we quote the specifications' own values, which the agreement on the calibration rows licenses. For the HQC-type schemes the message instance is the $[2n, n]$ syndrome instance of $u = r_1 + hr_2$ (weight $2w_r$), which is never harder than the full $[3n, n+k]$ instance; for QUBE it is the $[3r, 2r]$ instance of $u = g + h_1 f_1 + h_2 f_2$ (weight $w_g + 2w_f$); for Mito the 2-adic (resp. 4-adic) instances are $[4n, 2n]$ (resp. $[8n, 4n]$) with the summed block weights, and we also give the instance obtained by the augmentation $Z \mapsto 1$ of the group ring, which maps the 2-adic relation to a plain HQC relation of the same weight (§3.5); for BIKE-MLThre and TRIKE the key instance is codeword finding ($[2r, r]$, resp. $[3r, 2r]$) and the message instance is syndrome decoding ($[2r, r]$, resp. $[3r, r]$). The quasi-cyclic discount is $\frac{1}{2}\log_2 n$ (syndrome decoding) or $\log_2 n$ (codeword finding).

Three observations. (i) Within the Stern family the estimator's minimum is always Stern itself, and the May–Ozerov/BJMM gain over it is at most about two bits at these weights, so the ranking is robust to the algorithm choice. (ii) Where a specification publishes the same quantity, we reproduce it to the second decimal (QCTM's Table 1, TRIKE's Table 9; see §2), which is what licenses us to use the specifications' own numbers for the instances not recomputed here. The schemes differ in which discount they then apply—TRIKE and QCTM take the quasi-cyclic $\log_2 r$ or $\frac{1}{2}\log_2 \ell$, QUBE takes no quasi-cyclic discount and subtracts instead a "combinatorial gap" for its unbalanced block weights (8.10 bits on the key instance, 44.20 on the ciphertext instance)—so the column that matters for comparison is the raw one. (iii) HEP-QC-1/3/5 are HQC-1/3/5 with the targets relabelled from 143/207/272 to 128/192/256 and a public key 127× larger.

### 3.3 Same-key multi-instance decoding

Table 4 applies the DS-DOOM estimator [8] to the message instances. The published crossings (BIKE-MLThre-128/256, kem-05-2; HARE-256/ 384/512, kem-16-2; QCTM128/256, kem-32-2) are reproduced to within the dimension convention ($k = r$ versus $k = r - 1$).

One caveat on reading Table 4 against the specifications' own tables. At $M = n$ targets—one observed ciphertext with its $n$ free rotations—DS-DOOM is *more* expensive than the classical $\frac{1}{2}\log_2 n$ DOOM discount the designers apply: for TRIKE-1's message instance DS-DOOM gives 140.94 bits where the specification's Table 10 gives 132.11. The refined model is thus the conservative one at a single ciphertext, and the crossings below are crossings of the *designers' claimed level* by the *refined* estimate. New crossings, all with the estimator pinned by the tracker:

- **TRIKE.** The submitted sets TRIKE-5/7/9 cross their targets after $2^{47}$, $2^{63}$ and $2^{63}$ ciphertexts; TRIKE-2 (the submitted 128-bit set) stays above $2^{128}$ up to $2^{80}$. The NIST-level-1 set TRIKE-1 listed in the specification (not submitted) crosses after only $2^{31}$ ciphertexts and is at $2^{116}$ after $2^{64}$; the specification's own single-instance margin for it is 4.1 bits.
- **TriQ-KEM** crosses at all four levels ($2^{73}$, $2^{77}$, $2^{77}$, $2^{78}$ ciphertexts).
- **QUBE** crosses at the 128, 192, 256 and 384 levels ($2^{71}$, $2^{72}$, $2^{74}$, $2^{80}$); QUBE-512 stays above by 0.5 bits at $2^{80}$.
- **HARE-1** ($n = 16067$, the NIST-level-1 set) crosses at $2^{66}$; HARE-2 ($n = 20899$) does not.
- **NSS-HQC-128** crosses at $2^{77}$ (its larger sets are already below target with a single ciphertext, kem-26-3).
- **HEP-QC**, **Mito** and **BIKE-MLThre-512** do not cross.

The model is heuristic and several crossings are by fractions of a bit; the TRIKE-1 and TRIKE-5 crossings (12 and 0.3 bits at $2^{64}$ and $2^{47}$) and the TriQ/QUBE crossings at $2^{73}$–$2^{80}$ are of the same nature as the published ones for BIKE-MLThre and HARE. None of the specifications caps the number of encapsulations per key. The tracker has since filed these as kem-26-6 (NSS-HQC-128 same-key), kem-26-7 (NSS-HQC-384/512 static-key), kem-33-5 (QUBE), kem-36-9 (TRIKE) and kem-37-3 (TriQ-KEM), under the same estimator-and-key-mapping standard as the earlier kem-05-2/16-2/32-2. Their Confirmed status covers the stated *model* crossings, not a completed full-size decoding run; the crossings often exceed the level by less than a bit, and all assume many independently generated ciphertexts under one static key whose errors are unknown to the attacker. Two scope notes: QUBE-192 is assessed against QUBE's own claim, not a separate NGCC level; and HARE-1 and TRIKE-1/3 are the specifications' own small sets, not submitted implementation parameter sets.

### 3.4 Failure rates

QUBE's five values are reproduced to the printed precision, as are Mito1's three (from the specification's four-digit $p^*$; the 2-adic product model reproduces $p^*$ itself). TriQ-KEM's four *natural* values quoted below are its specification's own Table-6 bounds, not an output of our Model-1 script, which covers the HQC/QUBE concatenated structure and not TriQ-KEM's; we quote them as published. HEP-QC-7 has $2^{-574}$ under Model 1. Two observations follow.

**TriQ-KEM's failure rate misses a generic proof term, but meets the specification's own criterion.** The specification states natural failure rates $2^{-121.90}$, $2^{-173.35}$, $2^{-209.57}$, $2^{-230.00}$ for the 128, 256, 384, 512-bit sets (Table 6 of the specification); they are 6.1, 82.6, 174.4 and 282.0 bits short of a generic $2^{-\lambda}$. The specification replaces the standard requirement by an "attack-aware cumulative decryption-failure" criterion: the probability that *any* of $2^{80}$ adaptively chosen ciphertexts fails, after a bounded-density rejection of the encryption randomness, is bounded by $2^{-34.32}$, $2^{-85.63}$, $2^{-124.93}$, $2^{-146.56}$. Whatever one thinks of the criterion, the Fujisaki–Okamoto bound used in §V-A of the specification loses $q_D\delta$, which with $q_D = 2^{80}$ is $2^{-41.9}$, $2^{-93.4}$, $2^{-129.6}$, $2^{-150.0}$: the IND-CCA2 claim is not supported at any level by the proof the specification itself invokes. The specification's own attack-aware figure says the same thing in its own terms—a 128-

bit claim allows an adversary spending $2^{80}$ queries an advantage of at most $2^{80-128} = 2^{-48}$, and the post-rejection bound for TriQ-KEM-128 is $2^{-34.32}$, which is $2^{13.7}$ times larger. We stress the status of this observation: TriQ-KEM *meets* its specification's own stated criterion, and the shortfall is against a generic $q/2^{\lambda}$ proof term rather than a demonstrated failure attack. Under the tracker's classification policy a proof-term comparison of this kind is not a separate finding, and we do not claim one; we record it as a gap between the invoked FO bound and the stated margin, to be weighed alongside the criterion the designers chose. [CITED] (the figures are the specification's Table 6; the FO loss term is arithmetic on them).

**The others.** HARE's headline rates ($< 2^{-130}$, $< 2^{-258}$) rely on a Model-2 computation for which no artifact was submitted; the Model-1 values are $2^{-121.4}$ and $2^{-239.2}$ (kem-16-1, [CITED]). NSS-HQC's rate is simply wrong: 7 of 50,000 honest NSS-HQC-256 sessions failed (kem-26-2, [CITED]). TRIKE's and BIKE-MLThre's rates are extrapolations of bit-flipping simulations; TRIKE omits the boundary weak-key classes (kem-36-4, [CITED]), and BIKE-MLThre reports only iteration-wise statistics (0% failures at iteration 2 in limited experiments) for a learned threshold rule, which is no evidence for $2^{-128}$. Mito-E's distance-informed erasure decoding, whose failure analysis is the point of the E variants, is not implemented by the submitted code (kem-23-1, [CITED]).

### 3.5 Per-scheme notes and verdicts

**HARE (kem-16).** HQC with a covering-code compression of the second ciphertext component and an RM–RS decoder with erasures; six sets (Table 2 of the specification). Published: Model-2 DFR not reproducible (kem-16-1), multi-instance crossings at 256/384/512 (kem-16-2). Ours: crossings reproduced; HARE-1 crosses at $2^{66}$; single-instance margins (Table 3) are HQC-like. *Verdict: below the claimed level in the multi-ciphertext model (HARE-1/5/7/9); DFR unverified for HARE-128/256.*

**HEP-QC (kem-17).** HQC-1/3/5 parameters plus a 512-bit set, with the concatenated code's generator matrix $G$ replaced by a disguised $G' = T[G \mid GA]P$ (a random row mixing $T$, 64 extra random columns $GA$, a column permutation $P$; the inner RM(1, 7) code is duplicated $n_2/128$ times, so its columns come in classes of that multiplicity) and a second hardness assumption (EPC-P: $G'$ is indistinguishable from random). Xie's finding kem-17-4 falsifies EPC-P in linear time by column multiplicities ($3^{5888}1^{64}$ on every HEP-QC-1 key) and recovers $T$ and the affine blocks on 35/35 keys; Saarinen's kem-17-2 shows the 512-bit set is generated from a 256-bit seed. Our reading of the construction (the ciphertext is $v = mG' + \mathrm{Expand}(sr_2 + e)$ with $s$ dense) is that EPC-P is not needed for one-wayness: an attacker who knows $G'$'s structure still faces the HQC instance $u = r_1 + hr_2$. Falsifying EPC-P therefore breaks the *argument*—the specification bounds security by $\min(\text{EPC-P}, \text{QCSD})$ and the first term is now worthless—without breaking the *scheme*, whose one-wayness is HQC's. Since HQC-1/3/5 are worth 143/207/272 bits by the usual accounting while HEP-QC claims only 128/192/256 for the same parameters, those sets are if anything conservative; what the disguise buys is a public key 127× larger. HEP-QC-7 is a different matter: whatever its code parameters, its entire key pair comes from a 32-byte seed. *Verdict: EPC-P broken (previously), HEP-QC-7 below claim; HEP-QC-1/3/5 equal HQC with a* 127× *larger key and a claim lowered from* 143/207/272 *to* 128/192/256.

**NSS-HQC (kem-26).** HQC with a five-bit compressor, asymmetric error weights and RM–RS decoding. Published: DQCSD/JPCI falsified by parity (kem-26-1), DFR wrong and failure-oracle key recovery (kem-26-2, probable), ISD below target at 256/384/512 (kem-26-3: 226.5/333.0/441.4),

256-bit decryption seed at 384/512 (kem-26-4), RS-decoder timing key recovery at 128 (kem-26-5, probable). Ours: the single-instance message numbers are reproduced (Table 3); NSS-HQC-128 crosses at $2^{77}$. We add an independent shortfall on the *key*-recovery side. The specification prices key recovery not with an estimator but with an extrapolation formula $T_{\text{key}} = 1.0969\sqrt{n}$ (Tables 7–8), fit to a single HQC-128 data point, giving 188.2/256.1/384.0/512.0 for the four levels—exactly on target at 256 and above. A direct Stern estimate of the key instance $[2n, n]$ of weight $2w_{\text{sk}}$ gives 167.9/255.9/373.0/489.8 bits raw and 160.5/248.0/364.5/480.9 once the quasi-cyclic decoding discount the designers use elsewhere is applied ([COMPUTED], `nss_key.py`). Even before any discount the formula overstates the true cost (by 11 bits at 384, 22 at 512); with the discount the 256/384/512 key instances fall 8.0, 19.5 and 31.1 bits below their levels, the shortfall growing with $n$. The extrapolation is therefore not a security computation and is non-conservative, compounding the message-side break. *Verdict: previously broken/below claim at 256/384/512 (message), key extrapolation non-conservative by* 8–19 *bits at 256/384 (new, [COMPUTED]); 128 below claim in the multi-ciphertext model and subject to two probable attacks.*

**TriQ-KEM (kem-37).** HQC with bounded-density rejection sampling of $(r_1, r_2)$ and the attack-aware failure criterion. Published: two side-channel findings. Ours: failure rate above $2^{-\lambda}$ at every level (§3.4); multi-instance crossings at every level; single-instance margins of 18–20 bits over HQC-style accounting. *Verdict: not as secure as claimed (IND-CCA2 proof does not apply) and below claim in the multi-ciphertext model.*

**QUBE (kem-33).** Unbalanced multi-block QCSD: key $(3, 1)$-QCSD with weights $(w_x, w_x, w_y)$, ciphertext $(4, 2)$-QCSD, RM–RS decoding, Model-1 DFR (reproduced). Published: two side-channel and two implementation findings (optimised build only). Ours: QUBE's "combinatorial gap" is a correct discount *in the attacker's favour*—a generic estimator prices a weight-$w$ error in a search space of size $\binom{\sum n_i}{\sum w_i}$, whereas the real error lives in $\prod_i \binom{n_i}{w_i}$, which is smaller by the gap—so the designers' accounting is conservative where it applies, and their overall figures (136.5 for QUBE-128, against a requirement of 128) already carry margins of under ten bits. Our re-pricing of the ciphertext instance by the route that discards $v$ ($u = g + h_1 f_1 + h_2 f_2$, a $[3r, 2r]$ instance of weight $w_g + 2w_f = 85$) gives $154.43 \to 147.59$ after the standard $\frac{1}{2}\log_2 r$ discount: that particular route is not the binding one, which confirms rather than undercuts their choice. What their tables do not consider at all is the multi-ciphertext setting, where QUBE-128/192/256/384 cross their targets after $2^{71}$, $2^{72}$, $2^{74}$ and $2^{80}$ encapsulations (Table 4). *Verdict: below the claimed level in the multi-ciphertext model; single-instance margins already thin by the designers' own accounting.*

**Mito (kem-23).** HQC over the group ring $\mathcal{R}_n[\mathbb{Z}_2^e]$ ($e = 1$: Mito1, Mito1-E; $e = 2$: Mito2-E), i.e. 2-adic and 4-adic block structures, with RM–RS decoding (errors-only, or distance-informed erasures for the E variants).

**Proposition 1** (augmentation projection)**.** *Let $A = \mathcal{R}_n[Z_1, \ldots, Z_e]/(Z_1^2 - 1, \ldots, Z_e^2 - 1)$ be Mito's group ring ($e = 1$ for Mito1/Mito1-E, $e = 2$ for Mito2-E) and let $\varepsilon\colon A \to \mathcal{R}_n$ be the augmentation $Z_i \mapsto 1$. Then $\varepsilon$ is a surjective ring homomorphism, and the public relation $s = x + hy$ maps to $\varepsilon(s) = \varepsilon(x) + \varepsilon(h)\varepsilon(y)$ with $\mathrm{wt}(\varepsilon(x)) \le \sum_g \mathrm{wt}(x_g)$. Hence Mito key recovery is never harder than a plain HQC key-recovery instance of length $2n$, dimension $n$ and weight at most the total secret weight.*

*Proof.* $\varepsilon$ is the $\mathcal{R}_n$-algebra map induced by $Z_i \mapsto 1$, which is well defined since $1^2 - 1 = 0$; it is surjective because it fixes $\mathcal{R}_n$. The weight bound is the triangle inequality for the sum of the components, cancellations only helping. Applying $\varepsilon$ to the relation commutes with the ring operations. □

([Proved]; also [Measured] on 200 random instances for $e = 1, 2$ with $n = 127$, `mito_toy.py`, where cancellation lost 0.25 and 0.40 of a unit of weight on average.) The projection gives the attacker an HQC instance of half (resp. a quarter of) the length with the *same* total weight and half (resp. a quarter) of the rotations. Table 3 prices both; the projection gains about two bits (e.g. Mito1-128: 193.4 versus 195.6 under DS-DOOM), because ISD costs at fixed weight depend only weakly on the length. Mito's margins are therefore large (65 bits at the 128 level): the $2^\ell$-adic structure buys parallelism, not compactness (encapsulation key $2^\ell\lceil n/8\rceil + \lambda_{\text{seed}}$ bytes, i.e. 3908 for Mito1-128 and 5192 for Mito2-E-128, against HQC-128's 2241). We found no structural weakness beyond this; the only design-relevant finding is that the E variants' decoder is not the one implemented (kem-23-1). *Verdict: no design-level weakness found; E-variant failure claims unverified.*

**BIKE-MLThre (kem-05).** BIKE Level 1 and Level 5 parameters ($r = 12323$ and 40973) with a learned ("ML") threshold rule in the BGF decoder, plus a new 512-bit set ($r = 150001$, $w = 546$, $t = 524$). Published: unseeded default DRBG (kem-05-1, implementation), multi-instance crossings at 128/256 (kem-05-2). Ours: crossings reproduced; the 512-bit ring violates the 2-primitive rule (§3.1); the failure-rate evidence is iteration-wise only. *Verdict: BIKE's security at 128/256 (hence the same multi-ciphertext shortfall); 512 unverified in both DFR and ring structure.*

**TRIKE (kem-36).** A three-block QC-MDPC KEM: secret $(h_0, h_1, h_2)$ of total weight $w$, public key $(t_1, t_2, r_1, r_2)$ with $t_1, t_2, r_1$ random and $r_2 = t_2 + (r_1 + t_1)x$, where $x = (h_2 + t_2h_0)/(h_1 + t_1h_0)$ (Definition 9 of the specification). The ratio $x = (r_2 + t_2)/(r_1 + t_1)$ is therefore public, and $(xt_1 + t_2)h_0 + xh_1 + h_2 = 0$ exhibits $(h_0, h_1, h_2)$ as a weight-$w$ codeword of the public $[3r, 2r]$ code with parity-check $[\,xt_1 + t_2 \mid x \mid 1\,]$: key recovery is the 3-QCCF problem, exactly as the specification's reductions (Theorems 1–2) say, while a ciphertext $(u, v) = (e_0 + r_1e_1 + r_2e_2,\ e_0 + t_1e_1 + t_2e_2)$ is a $(2, 3)$-QCSD instance ($[3r, r]$, weight $t$). Published: the specified decoder rejects all honest ciphertexts (kem-36-1), weak-key classes omitted from the DFR bound (kem-36-4), Theorem 4's reduction uses secret data (kem-36-5), side-channel and memory findings. Ours: the designers' ISD tables are reproduced; the multi-instance crossings of §3.3. *Verdict: below the claimed level in the multi-ciphertext model (TRIKE-5/7/9; TRIKE-1 dramatically); correctness claims unproved.*

## 4 QCTM: the twist is cosmetic and the fold lifts

QCTM (kem-32) is a Niederreiter-type KEM whose public code is presented as a *quasi-cyclic twisted Goppa code*. Its three parameter sets are $(m, \ell, t_0, n) = (18, 19, 15, 10070)$, $(18, 19, 27, 19000)$ and $(18, 19, 53, 38000)$, with Goppa degree $t = \ell t_0$, error weight $w = (t - 1)/2$ and dimension $k = n - m(t - 1) - 1$. The specification's structural argument (§6.3 of the specification) is that the twist destroys the hypotheses of the folding attack of Faugère, Otmani, Perret, de Portzamparc and Tillich [12], so that the quotient code is "neither a GRS nor a twisted GRS code". The forum has already established the opposite empirically: L. Wang observed (kem-32-4, audited by Saarinen on the archived KAT keys) that folding the public matrix across the orbits yields the parity-check space of an ordinary binary Goppa code with one parity row, and gave quotient-only cost estimates (kem-32-5) under the heuristic GIJS cost model. The tracker lists three gaps: the quotient attack is not established at these degrees, no real quotient key has been recovered, and "lifting a quotient key through the order-19 alignment and twist gauge to a full decapsulation key has not been implemented or costed".

This section closes the third gap and gives the first proof of the first observation. We prove (i) that the twisted row is a linear combination of the ordinary rows and the all-ones vector, so that the public code is an *alternant* code intersected with the even-weight code and corrects only half as many errors as a Goppa code of the same redundancy; (ii) that the folded code is *exactly* the binary Goppa code $\Gamma(X', M) \cap E$ with $X' = \{a_O^\ell\}$; and (iii) that any quotient key, in any of the equivalent representations a Goppa key-recovery algorithm may return, lifts in polynomial time to a decoding key for the full QCTM code. All three statements are verified numerically on scaled-down instances that follow the specification's key generation literally (Algorithms 1, 3, 4). Consequently the key security of QCTM is *polynomially equivalent* to the key security of a binary Goppa code with parameters $(m, n/\ell, t/\ell) = (18, 530, 15)$, $(18, 1000, 27)$, $(18, 2000, 53)$, whose support lies inside the index-19 subgroup of $\ell$-th powers.

### 4.1 The construction

Let $q = 2^m$, $\mathbb{F} = \mathbb{F}_q$, $\ell$ an odd prime with $\ell \mid q-1$, $\zeta \in \mathbb{F}$ of order $\ell$ (a public parameter), $t_0$ odd and $t = \ell t_0$. Key generation samples $\eta \in \mathbb{F}^*$ and a monic irreducible $M \in \mathbb{F}[y]$ of degree $t_0$ such that $M(y^\ell)$ is irreducible, sets

$$g(x) = M\big((x - \eta^{-1})^\ell\big) = \sum_{i=0}^{t} g_i x^i, \qquad g_t = 1,$$

and chooses $n_b = n/\ell$ nonzero elements $a_1, \ldots, a_{n_b}$ with pairwise distinct orbits $\{\zeta^i a_O\}_{0 \le i < \ell}$. The support is the concatenation of the affine orbits $L = (\zeta^i a_O + \eta^{-1})_{O,i}$ (Algorithm 3). The $t \times n$ matrix $H_T$ of Algorithm 4 has rows $\alpha^i/g(\alpha)$ for $0 \le i \le t-2$ and the *twisted* row $(\alpha^{t-1} + \eta\alpha^t)/g(\alpha)$, evaluated at $\alpha \in L$; the first $t-1$ rows are expanded over $\mathbb{F}_2$ and the last row is replaced by the all-ones row (the specification states that "after elementary row operations the last row is an all-one row"). The public key is a hybrid systematic form of the resulting $u \times n$ binary matrix, $u = m(t-1)+1$, with the $\ell$-block-circulant part stored by its first rows; in particular the orbit partition of the columns and the cyclic order inside every orbit are public. The ciphertext is the syndrome $He^\top$ of a weight-$w$ error.

### 4.2 The twisted row is a combination of the ordinary rows

**Lemma 2.** $g_{t-1} = \eta^{-1}$.

*Proof.* Write $g(x) = \sum_{s=0}^{t_0} M_s (x - \eta^{-1})^{\ell s}$ with $M_{t_0} = 1$. The terms with $s < t_0$ have degree at most $\ell(t_0 - 1) = t - \ell \le t - 3$. The coefficient of $x^{t-1}$ in $(x - \eta^{-1})^t$ is $\binom{t}{1}(-\eta^{-1}) = t\eta^{-1} = \eta^{-1}$ because $t$ is odd and the characteristic is 2. □

**Lemma 3** (cosmetic twist)**.** *For every $\alpha$ with $g(\alpha) \neq 0$,*

$$\frac{\alpha^{t-1} + \eta\alpha^t}{g(\alpha)} \;=\; \eta \;+\; \eta \sum_{i=0}^{t-2} g_i \, \frac{\alpha^i}{g(\alpha)}.$$

*Proof.* $\eta\alpha^t = \eta g(\alpha) - \eta \sum_{i<t} g_i \alpha^i$, hence $(\alpha^{t-1} + \eta\alpha^t)/g(\alpha) = \eta + \big((1 - \eta g_{t-1})\alpha^{t-1} - \eta \sum_{i \le t-2} g_i \alpha^i\big)/g(\alpha)$, and $1 - \eta g_{t-1} = 0$ by Lemma 2; signs are irrelevant in characteristic 2. □

The twisted row therefore adds nothing beyond the all-ones vector (which the specification keeps explicitly), and the $\mathbb{F}$-row space of $H_T$ is $\mathrm{span}\{\alpha^i/g(\alpha)\}_{i \le t-2} \oplus \langle \mathbf{1} \rangle$.

**Theorem 4** (the public code). *Let $\mathcal{A}_r(L,\mathbf{y}) = \{c \in \mathbb{F}_2^n : \sum_j c_j y_j \alpha_j^i = 0,\ 0 \le i < r\}$ denote the binary alternant code of order $r$ and let $E$ be the even-weight code. The QCTM public code is*

$$\mathcal{C}_{\mathrm{pub}} = \mathcal{A}_{t-1}\big(L, g^{-1}\big) \cap E,$$

*of dimension $n - m(t-1) - 1$ whenever the binary expansion has full rank (the specification's assumption). Its minimum distance is at least $t+1$, and the specification's decoder corrects $w = (t-1)/2$ errors, which is the generic alternant radius. In particular $\mathcal{C}_{\mathrm{pub}}$ is not a Goppa code: the binary Goppa code $\Gamma(L,g) = \mathcal{A}_t(L,g^{-1})$ with the same support and polynomial would correct $t$ errors with redundancy $mt = m(t-1)+m$, so QCTM spends essentially the same redundancy for half the error weight.*

*Proof.* By Lemma 3 the parity checks are $\sum_j c_j \alpha_j^i / g(\alpha_j) = 0$ for $i \le t-2$ together with $\sum_j c_j = 0$. The first family defines $\mathcal{A}_{t-1}(L,g^{-1})$; an alternant code of order $r$ has designed distance $r+1$, and intersecting with $E$ raises an even designed distance $t$ (with $t$ odd) to $t+1$. The last two claims are the classical facts on Goppa codes with square-free $g$ ($\Gamma(L,g) = \Gamma(L,g^2)$). □

### 4.3 The quasi-cyclic action and the fold

Put $u_j = \alpha_j - \eta^{-1}$, so that $u_{(O,i)} = \zeta^i a_O$ and $g(\alpha_j) = M(u_j^\ell)$. Because $\{\alpha^i\}_{i \le t-2}$ and $\{u^i\}_{i \le t-2}$ span the same polynomial space (triangular change of basis), Theorem 4 gives

$$\mathcal{C}_{\mathrm{pub}} = \mathcal{A}_{t-1}(U,\mathbf{y}) \cap E, \qquad U = (\zeta^i a_O)_{O,i}, \quad y_{(O,i)} = 1/M(a_O^\ell) =: y_O. \tag{1}$$

The multiplier is constant on orbits and the orbit of $u$ under $\sigma\colon (O,i) \mapsto (O,i+1)$ is $u \mapsto \zeta u$, so each parity row $u^i y$ is mapped to $\zeta^i \cdot (u^i y)$ and $\sigma$ is an automorphism of $\mathcal{C}_{\mathrm{pub}}$; the secret $\eta$ has disappeared from the description of the code. The *folded code* is $\varphi(\mathcal{C}_{\mathrm{pub}}) = \{(\sum_i c_{(O,i)})_O : c \in \mathcal{C}_{\mathrm{pub}}\} \subseteq \mathbb{F}_2^{n_b}$.

**Theorem 5** (the fold is a Goppa code). *Let $X' = (x_O)_O$ with $x_O = a_O^\ell$ and let $M$ be the degree-$t_0$ polynomial of the key. Then*

$$\varphi(\mathcal{C}_{\mathrm{pub}}) = \Gamma(X', M) \cap E,$$

*the binary Goppa code of length $n_b$ with (irreducible) Goppa polynomial $M$ and support $X' \subseteq (\mathbb{F}^*)^\ell$, intersected with the even-weight code. Its dimension is at least $n_b - mt_0 - 1$ (equality on all tested keys and on the archived KAT keys audited in kem-32-4: $[530, 259]$, $[1000, 513]$, $[2000, 1045]$).*

*Proof.* Since $\ell$ is odd, $\pi = \sum_{k=0}^{\ell-1} \sigma^k$ maps $\mathcal{C}_{\mathrm{pub}}$ onto its $\sigma$-invariant subcode (for an invariant $c$, $\pi(c) = \ell c = c$), and $\varphi$ is the composition of $\pi$ with the compression that keeps one coordinate per orbit. A vector $c$ constant on orbits ($c_{(O,i)} = \bar{c}_O$) lies in $\mathcal{C}_{\mathrm{pub}}$ iff for all $0 \le i \le t-2$

$$0 = \sum_O \bar{c}_O\, y_O \sum_{k=0}^{\ell-1} (\zeta^k a_O)^i = \sum_O \bar{c}_O\, y_O\, a_O^i \sum_k \zeta^{ki} = \begin{cases} \ell \sum_O \bar{c}_O\, y_O\, x_O^{i/\ell} & \ell \mid i, \\ 0 & \text{otherwise,} \end{cases}$$

and $\sum_O \ell \bar{c}_O = \sum_O \bar{c}_O = 0$. The surviving exponents are $i = \ell i'$ with $0 \le i' \le \lfloor (t-2)/\ell \rfloor = t_0 - 1$, i.e. the checks $\sum_O \bar{c}_O x_O^{i'} / M(x_O) = 0$ for $i' < t_0$, which define $\mathcal{A}_{t_0}(X', M^{-1}) = \Gamma(X', M)$. □

Theorem 5 is the exact form, for QCTM, of the general "folding is alternant" theorem of [12]; the specification's counter-argument fails because the twist never enters the public code. Note also that the fold has one further invariant: its support consists of $\ell$-th powers, so it is a Goppa code whose support is confined to the multiplicative subgroup of index 19 (of order 13797 for $m = 18$).

### 4.4 Lifting a quotient key

Suppose a key-recovery algorithm for binary Goppa codes returns a pair $(X'', M'')$ with $\Gamma(X'', M'') \cap E = \varphi(\mathcal{C}_{\mathrm{pub}})$. Such a pair is determined only up to the usual equivalences: translations and scalings of the support, Frobenius conjugation, and scaling of $M$; we may therefore assume $X'' = \big(\alpha_0 X' + \beta_0\big)^{2^{j_0}}$ for unknown $\alpha_0 \in \mathbb{F}^*, \beta_0 \in \mathbb{F}, j_0 \in [0, m)$, with the matching $M''$. The lifting problem is to recover a description of $\mathcal{C}_{\mathrm{pub}}$ itself, i.e. the per-orbit roots $a_O$, correctly aligned with the public cyclic order inside each orbit. Scaling $U \mapsto \lambda U$ and the global rotation $\sigma$ are harmless (they do not change the code), so the real unknown is the vector of *rotations*: choosing any $b_O$ with $b_O^\ell = x_O$, the true roots are $a_O = \rho_O b_O$ with $\rho_O \in \mu_\ell$, and different $\rho$ give different codes.

**Character projections of the public dual.** Let $D = \mathrm{GRS}_{t-1}(U, \mathbf{y}) + \langle \mathbf{1} \rangle \subseteq \mathbb{F}^n$ be the $\mathbb{F}$-row space of the parity-check rows. By Delsarte's theorem the public dual code is $\mathcal{C}_{\mathrm{pub}}^\perp = \mathrm{Tr}(D)$ and $\widetilde{D} := \mathcal{C}_{\mathrm{pub}}^\perp \otimes \mathbb{F} = \sum_{k=0}^{m-1} D^{(2^k)}$ (sum of Frobenius conjugates), a space the attacker can compute from the public matrix. For $\chi \in \mathbb{Z}_\ell$ define the block-wise character projection $P_\chi \colon \mathbb{F}^n \to \mathbb{F}^{n_b}$, $(P_\chi v)_O = \sum_i \zeta^{-i\chi} v_{(O,i)}$ (with the public $\zeta$ and the public in-orbit order).

**Lemma 6.** *Let $e_k = 2^{-k} \bmod \ell$ and $d_k = \lfloor (t - 2 - e_k)/\ell \rfloor$ for $0 \le k < m$, and write $\mathbf{a} = (a_O)_O$, $\mathbf{x} = (x_O)_O$. Then*

$$P_1(\widetilde{D}) = \mathbf{a} \star E_1, \qquad E_1 = \sum_{k=0}^{m-1} \mathbf{x}^{(e_k 2^k - 1)/\ell}\, \mathbf{y}^{2^k} \star \mathrm{GRS}_{d_k+1}\big(\mathbf{x}^{2^k}\big),$$

*where $\star$ is the coordinate-wise product and $\mathrm{GRS}_d(\mathbf{z}) = \{(G(z_O))_O : \deg G < d\}$. The space $E_1$ depends only on the quotient data $(\mathbf{x}, \mathbf{y})$, and $\dim_{\mathbb{F}} P_1(\widetilde{D}) \le \sum_k (d_k + 1)$.*

*Proof.* Take $v = f(u)^{2^k} y^{2^k} \in D^{(2^k)}$ with $\deg f \le t - 2$. On orbit $O$, $v_{(O,i)} = \sum_j f_j^{2^k} a_O^{j 2^k} \zeta^{ij2^k} y_O^{2^k}$, so (using $\ell \equiv 1$ in characteristic 2) the character-1 component keeps the $j$ with $j 2^k \equiv 1$, i.e. $j = e_k + \ell h$, $0 \le h \le d_k$: $(P_1 v)_O = a_O^{e_k 2^k} y_O^{2^k} G(x_O^{2^k})$ with $G = \sum_h f_{e_k+\ell h}^{2^k} Y^h$ arbitrary of degree $\le d_k$. Finally $a_O^{e_k 2^k} = a_O\, x_O^{(e_k 2^k - 1)/\ell}$ because $e_k 2^k \equiv 1 \pmod{\ell}$ and $e_k 2^k \ge 1$. The all-ones vector has zero character-1 component since $\sum_i \zeta^{-i} = 0$. □

**The lifting algorithm.**

1. *Normalisation.* Find $\beta$ such that all $x''_O - \beta$ lie in one coset of the subgroup $(\mathbb{F}^*)^\ell$ (test $(x''_O - \beta)/(x''_1 - \beta)$ for $\ell$-th-power-ness; $q$ candidates, $n_b$ tests each). Set $\mathbf{x}_{\mathrm{att}} = (X'' - \beta)/(x''_1 - \beta) \subseteq (\mathbb{F}^*)^\ell$ and $\mathbf{y}_{\mathrm{att}} = 1/M''(X'')$; the Goppa code is unchanged by the affine renormalisation. A random translate is a single coset with probability $\approx \ell^{-(n_b - 1)}$, so $\beta$ is unique in practice.
2. For each Frobenius hypothesis $j \in [0, m)$: replace $(\mathbf{x}, \mathbf{y})$ by their $2^{-j}$-th powers, pick any $\mathbf{b}$ with $\mathbf{b}^\ell = \mathbf{x}$, and form $K_1 = P_1(\widetilde{D})$ from the public matrix and $K_2 = \mathbf{b} \star E_1(\mathbf{x}, \mathbf{y})$ from Lemma 6.
3. Solve the *linear* system $\rho \star K_2 \subseteq K_1$, i.e. $\sum_O \rho_O v_O w_O = 0$ for all $v$ in a basis of $K_2$ and $w$ in a basis of $K_1^\perp$ ($\dim K_2 \cdot \dim K_1^\perp$ equations in $n_b$ unknowns). Accept $j$ iff the solution space is one-dimensional and, after normalising $\rho_1 = 1$, all $\rho_O$ are $\ell$-th roots of unity.
4. Output $U^* = (\zeta^i \rho_O b_O)_{O,i}$ and $\mathbf{y}^* = (y_O)_{O,i}$. (If the quotient key is returned with a Goppa polynomial that is a Frobenius power of $M''$—e.g. $M''^2$, which defines the same binary code—the loop runs over independent Frobenius exponents for $\mathbf{x}$ and $\mathbf{y}$, $m^2$ hypotheses instead of $m$.)

5. *Decoding.* Build the binary parity-check matrix $\widehat{H}^*$ of $\mathcal{A}_{t-1}(U^*, \mathbf{y}^*) \cap E$, solve $H_{\text{pub}} = Q'\widehat{H}^*$ for the (public, invertible) change of basis $Q'$, map a ciphertext $s = H_{\text{pub}}e^\top$ to the $\mathbb{F}$-syndromes $S_i = \sum_j e_j y_j^* u_j^{*i}$ $(i \le t-2 = 2w-1)$ and run Berlekamp–Massey and a Chien search; $2w$ syndromes locate $w$ errors.

**Theorem 7** (lifting)**.** *For the true $(j, \rho)$ the vector $\rho$ is a solution of step 3, and if the solution space is one-dimensional the output of step 4 satisfies $\mathcal{A}_{t-1}(U^*, \mathbf{y}^*) \cap E = \mathcal{C}_{\text{pub}}$; step 5 then decodes every honest ciphertext. The whole procedure is polynomial: the normalisation costs $O(q\, n_b)$ field operations, and each of the $m$ Frobenius hypotheses costs one projection of the $u$ public rows, one kernel computation on a $u \times n_b$ matrix and one linear system with $n_b$ unknowns and at most $(mt_0)^2$ equations, i.e. $O\big(m\,(u\,n + u\,n_b^2 + (mt_0)^2 n_b^2)\big)$ in all—negligible next to any quotient key recovery.*

*Proof.* Correctness of the true solution is Lemma 6 together with (1): for the correct Frobenius hypothesis the attacker's $(\mathbf{x}, \mathbf{y})$ equal the true $(\mathbf{x}/x_1, \mathbf{y})$ up to the harmless scaling, and $\mathbf{a} = \rho \star \mathbf{b}$. Uniqueness is a heuristic (the diagonal stabiliser of $E_1$ is trivial); it held in every experiment, and a one-dimensional solution space is *certified* by the final check $\mathcal{A}_{t-1}(U^*, \mathbf{y}^*) \cap E = \mathcal{C}_{\text{pub}}$, which is a rank computation over $\mathbb{F}_2$. The alternant decoder with $t-1 = 2w$ syndromes corrects $w$ errors. □

**Experiments.** The script `qctm_toy.py` generates keys exactly as in Algorithms 1, 3, 4 (irreducibility of $M$ and of $g$ tested by Ben-Or), hides the key behind a random invertible row mixing and a random block permutation, hands the attacker only the public matrix and a quotient key re-normalised by a random affine map and a random Frobenius twist, and runs the lifting. Table 6 lists the instances. [Proved] for Lemmas 2–3 and Theorems 4–5; [Proved] for the correctness half of Theorem 7, with the uniqueness of the solution in step 3 a heuristic that each run certifies for itself; [Measured] throughout for the experiments. In every run (i) the twisted row reduced to $\eta\mathbf{1}$ plus ordinary rows, (ii) $\sigma(\mathcal{C}_{\text{pub}}) = \mathcal{C}_{\text{pub}}$, (iii) $\varphi(\mathcal{C}_{\text{pub}}) = \Gamma(X', M) \cap E$, (iv) the normalising translation $\beta$ was unique, exactly one Frobenius hypothesis gave a one-dimensional solution with entries in $\mu_\ell$, $\dim K_1$ equalled $\sum_k (d_k + 1)$ as predicted, the lifted alternant key reproduced the public code, and a random weight-$w$ error was decoded from its public syndrome.

### 4.5 Consequences for the three parameter sets

**Corollary 8.** *Up to a polynomial-time reduction in both directions, recovering a QCTM decapsulation key is the key-recovery problem for a binary Goppa code $\Gamma(X', M)$ with $(m, n_b, t_0) = (18, 530, 15)$, $(18, 1000, 27)$ or $(18, 2000, 53)$, irreducible $M$, and support contained in the index-19 subgroup of $\mathbb{F}_{2^{18}}^*$. The reduction from QCTM to the quotient is Theorem 7; the reduction from the quotient to QCTM is trivial (a QCTM key gives the quotient key).*

The corollary is insensitive to the parameter revision the QCTM team announced on 2026-10-08 in response to kem-32-2 ($n = 10070 \to 10545$ and $19000 \to 19551$, with $m$, $\ell$ and $t_0$ unchanged). Both new lengths are again divisible by $\ell = 19$—they have to be, since the support is a union of 19-element orbits—so the construction, Lemma 3, Theorem 5 and the lifting all go through verbatim, and the only effect is to move the quotient from $(18, 530, 15)$ to $(18, 555, 15)$ and from $(18, 1000, 27)$ to $(18, 1029, 27)$. The revision addresses the multi-instance crossing and leaves the structural reduction exactly where it was.

Three remarks follow. First, the design goal of the twist is not met: the public code is a quasi-cyclic alternant code to which the structural attack framework of [12] applies verbatim, and the only thing that protects the key is the hardness of recovering the quotient Goppa code. Second, the quotient codes are far outside the regime of the classical high-rate distinguishers. QCTM128's

quotient is a $[530, 259]$ code, i.e. rate 0.489 with $n_b/(mt_0) = 1.96$; on toy quotients built exactly as in Theorem 5 (support inside the subgroup of $\ell$-th powers) the Schur square of the dual fills the whole space at every rate we could reach —$(m, \ell, t_0, n_b) = (10, 11, 3, 48)$, $(10, 11, 5, 80)$, $(10, 11, 7, 93)$, $(12, 13, 5, 160)$, $(12, 13, 7, 170)$, $(12, 13, 9, 280)$, with rates 0.24 to 0.62 and $n_b/(mt_0)$ from 1.1 to 2.7—so the quotient behaves like a random code for the distinguisher of [13] and its key-recovery descendants. [MEASURED] (`square_toy.py`). Third, the 2026 developments on Goppa key recovery (the higher-order-vanishing distinguisher of Hemmert and Wiemers [14, 15], the quasi-polynomial hold-out distinguisher of Ghoshal, Ishai, Jain and Sun [16], and the key-recovery routes of Vedenev [17], Weis [19] and Saarinen [20]) have so far been costed only for $n \approx 2^m$; L. Wang's estimates (kem-32-5) under Weis's heuristic model give $2^{141.3}$ and $2^{142.0}$ bit operations for the QCTM256 and QCTM512 quotients (with $2^{176.7}$, $2^{177.7}$ under square-root memory charging), i.e. far below the claimed 256 and 512 bits if the model's assumptions hold at $(18, 1000, 27)$ and $(18, 2000, 53)$; no estimate for the QCTM128 quotient has been published and we do not venture one. By Corollary 8 every such quotient estimate is, up to the cost of Theorem 7, an estimate for QCTM itself.

It is worth noting which way the one published *negative* result cuts here. Apon [18] argues that Vedenev's key-recovery route needs more than $2t+3$ held positions, which for mceliece8192128 ($t = 128$) means more than 259 and pushes the cost above $2^{1500}$; that bound is what currently protects Classic McEliece's largest set from this particular route. The same bound at QCTM's quotient degrees reads $2t_0 + 3 = 33$, 57 and 109. We have not re-derived Apon's argument and we do not translate its exponent into a cost, but the obstruction it describes is an increasing function of the Goppa degree, and QCTM's quotients have the smallest degrees of any Goppa instance currently in a submission. Whatever protects Classic McEliece here protects QCTM considerably less. We therefore classify QCTM256 and QCTM512 as *not as secure as claimed* (conditionally on the heuristic quotient estimates, and unconditionally for the multi-instance crossings of §3.3 and the RNG cap kem-32-3), and QCTM128 as *security open*: its message security is below $2^{128}$ in the multi-ciphertext model (kem-32-2 and Table 4) and its key security is that of a $(18, 530, 15)$ Goppa code with subgroup support, a problem for which neither an attack nor a hardness argument is known.

# 5 The rank-metric family

Five candidates are rank-metric KEMs in the RQC/LRPC lineage with structured errors: BRQC and BRA use *blockwise* errors (the supports of the blocks of the error are in direct sum) over ideal codes, with Gabidulin (BRQC) or augmented Gabidulin (BRA) decoding; C-Multi-UR-AG uses *multi-syndrome* (rank-support-learning) errors in unstructured codes; BAG-Loong and BAG-Piglet use block errors with, respectively, intersecting and disjoint supports and augmented Gabidulin decoding. All five submissions derive their security claims from the MaxMinors (MM) algebraic attack, priced with the linear-algebra exponent $\omega = 2$, which the designers call conservative.

## 5.1 Published status

BAG-Piglet is broken at the design level: its rejection key is derived from the re-encryption of the *decrypted* message rather than from the received ciphertext, which turns the decapsulation oracle into a plaintext-checking oracle; Saarinen recovers the decryption secret $x$ of BAG-Piglet-128 with 57,057 queries in five minutes and of BAG-Piglet-256 with $\approx 3.7 \cdot 10^5$ queries (kem-04-2, Critical/Confirmed, [CITED]). BAG-Loong has the same rejection-key design (kem-03-2), which the tracker lists only as Medium/design concern (reclassified down from High on 17 October 2026) and against which no key recovery had been run. We ran it: §5.4.1 reports an executed demonstration,

on BAG-Loong's own reference implementation, that the implicit-rejection shared secret does not bind the received ciphertext and leaks message equality ([Measured]). Separately, its reference sampler uses *public* support spaces so that the secret $X$ is recovered from the public key by linear algebra in under a second (Liu, kem-03-3, ePrint 2026/2223, [Cited]), and its security analysis prices the error rank as $t_1 + t_2$ where the two secret supports share a $g$-dimensional intersection, so that the true rank is $t_1 + t_2 - g$ (Xiong–Wang, kem-03-4, [Cited]). BRQC, BRA and C-Multi-UR-AG have only implementation-level findings on the tracker: decoder crashes on malformed inputs, unchecked padding bits that make ciphertexts malleable without changing the key (a one-query IND-CCA2 violation of the submitted *code*, not of the specified scheme: kem-06-4, kem-07-2, kem-10-3), secret-dependent memory addresses in the decoders with modelled key-recovery routes (kem-06-3, kem-07-1, leads), and DRBG misuse.

### 5.2 Blockwise MaxMinors: an independent re-implementation

The BRQC and BRA specifications bound the number of independent MM equations for blockwise errors by the function $\overline{E}(m, n, k, r, \ell, \mathbf{n}, \mathbf{r})$ of their Theorem 7.6 and price the attack, with blockwise puncturing $\mathbf{p}$, hybrid guessing $\mathbf{a}$ and sub-support guessing $\mathbf{h}$, by Theorem 7.7:

$$q^{\sum_i a_i(r_i-h_i)+h_i(n_i-h_i)} \cdot \overline{E} \cdot \Big(\prod_i \tbinom{n_i-p_i-a_i-h_i}{r_i-h_i}\Big)^{\omega-1},$$

subject to the feasibility criterion $\xi' := \overline{E}/N \geq 1$, where $N = \prod_i \binom{n_i-p_i-a_i-h_i}{r_i-h_i}$ is the number of unknowns.

We transcribed the two-case formula for $\overline{E}$ and validated it twice. In the homogeneous case ($\ell = 1$) it reduces to $m\binom{n-k-1}{r}$, the classical MM equation count, exactly. On the designers' own data points it reproduces their numbers:

| Instance | blocks $\mathbf{n}$ | ranks $\mathbf{r}$ | $(\mathbf{p};\mathbf{a})$ | spec ($\xi'$) | ours ($\xi'$) |
|---|---|---|---|---|---|
| BRQC-128 key | $(119, 119)$ | $(5, 5)$ | $(\mathbf{0}; 0, 22)$ | $2^{216}$ (1.05) | $2^{216.84}$ (1.012) |
| BRQC-128 ct | $(119, 119, 119)$ | $(5, 5, 8)$ | $(0, 0, 32; \mathbf{0})$ | $2^{181}$ (1.01) | $2^{181.32}$ (1.009) |
| BRQC-512 ct | $(227, 227, 227)$ | $(8, 8, 8)$ | $(\mathbf{0}; 0, 0, 36)$ | $2^{566}$ (1.03) | $2^{566.79}$ (1.031) |
| BRA-128 ct | $(121, 121, 121)$ | $(8, 5, 5)$ | $(1, 1, 2; \mathbf{0})$ | $2^{189}$ (1.01) | $2^{189.15}$ (1.014) |

The last line shows that the block *order* matters—the counting of Theorem 7.6 is not symmetric in the blocks, and the designers order them with the largest rank first. We also added the cap $\overline{E} \leq m\binom{\sum_i n'_i - k' - 1}{\sum_i r'_i}$, since the modelling has no more equations than that in total. [Computed] (`mm_blockwise.py`).

We then searched independently over block orderings, $\mathbf{p}$ (greedy column-by-column puncturing followed by a local exhaustive refinement), $\mathbf{a}$ (up to two non-zero blocks, each up to 90) and single-block $\mathbf{h}$, with $\omega = 2$ as the designers. An earlier version of this paper reported that this search found points below the designers' tabulated minima—by a few bits on several ciphertext instances, and, under aggressive per-block puncturing, by up to 150 bits. That conclusion was *wrong*, and we withdraw it. As Song, Chen, Zhang, Huang, Weng and Wang [26] establish, $\overline{E}$ is a *tight* upper bound on the number of independent MM equations only in the designers' block order (largest rank first) and away from the degenerate heavy-puncturing regime ($n'_i$ close to $r_i$); evaluated over free orderings or under heavy puncturing it overstates the independent equations, so a point with $\xi' = \overline{E}/N \geq 1$ there is not actually solvable. Our search had minimised over exactly those invalid regions, so its sub-tabulated points—including the $2^{542.6}$ and the $2^{414.1}$ reported before—are artifacts, not attacks. Restricted to the regime in which [26] proves $\overline{E}$ tight, our independent re-implementation reproduces

the designers' Table-6 values (as the homogeneous check and the reproduction of their published data points above already confirm), and finds no point below them. BRQC's and BRA's published strengths (181/295/566 and 189/304/556) therefore stand, against levels of 128/256/512: margins of 53 bits or more.

The *homogeneous* relaxation (an error of rank $\sum_i r_i$ with no block structure) under CryptographicEstimators, which is hundreds of bits higher: the blockwise structure is what makes these parameters small, and the designers' blockwise analysis is the binding one. One further caveat: BRA uses $m < n$ ($67 < 121$, $83 < 161$, $127 < 221$), so the matrix form of the problem is "wide"; the MM modelling used by the designers is the $\mathbb{F}_{q^m}$-linear one and we did not evaluate the transposed (MinRank) formulation, which for blockwise errors has no published analysis. We also note that CryptographicEstimators' MinRank "BigK" complexities are not meaningful in the regime $K \gg (m-r)(n-r)$; the minima we report in Table 7 are MaxMinors, not BigK MinRank, and are unaffected.

### 5.3 C-Multi-UR-AG: rank support learning

C-Multi-UR-AG publishes $N_1$ syndromes of errors sharing one support of dimension $w_1$ (the RSL problem $\mathrm{RSL}_{q,m,2n,n,w_1,N_1}$) and its ciphertexts contain $N_2$ syndromes sharing a support of dimension $w_2$. The specification prices the underlying RSD instances with MM ($\omega = 2$) at $2^{170}/2^{276}/2^{530}$ and asserts that algebraic attacks do not benefit from the multiple syndromes. Table 7 gives the same RSD instances under CryptographicEstimators with $\omega = 2.81$: 197/200 (key/ciphertext) at level 128, 327/313 at 256, 571/628 at 512, with the same optimal hybrid parameters ($a = 14$ and 11 at level 128) as the specification—the difference is $\omega$ alone. [Computed] (`rank_pricing.py`).

For the RSL aspect we tested the naive algebraic use of the extra syndromes: stacking the MM equations of all $N$ instances in the common unknowns $\{c_T\}$ (the maximal minors of the common coefficient matrix). On toy instances the rank of the stacked system saturates well below the number of unknowns: for $(m, n, k, r) = (10, 16, 8, 4)$ the ranks for $N = 1, \ldots, 6$ are $350, 550, 650, 690, 690, 690$ against 1819 unknowns, and $690 = m\binom{n-k}{r} - m$; for $(11, 14, 7, 3)$ they are $220, 330, 363$ (full) with $m\binom{n-k}{r} - m = 374 \geq 363$. [Measured] (`rsl_toy.py`). The reason is that every instance's equations are $r \times r$ minors of $(CH^\top)P_i^\top$ and hence lie in the span of the $\binom{n-k}{r}$ minors of the fixed matrix $CH^\top$; the multi-syndrome structure can therefore raise the equation count by at most the factor $\binom{n-k}{r}/\binom{n-k-1}{r} = (n-k)/(n-k-r)$, i.e. 0.4 bits at level 128. The genuinely stronger RSL attack of Bardet and Briaud [24] becomes polynomial only for $N$ comparable to $kr$ (280 at level 128 versus $N_1 = 11$); its algebraic refinement (journal version, IEEE Trans. Inf. Theory 2023) is already the basis of the specification's RSL analysis, so there is no un-evaluated intermediate regime here, and the combinatorial RSL gain is the $N$ bits the specification already accounts for. *Verdict: no design-level weakness found; the claimed levels hold with $\omega = 2$ margins of 42–98 bits under $\omega = 2.81$.*

### 5.4 BAG-Loong and BAG-Piglet

With the merged ranks $t_1 + t_2 - g = 6, 8, 9, 10$ of kem-03-4, the BAG-Loong key instances $[2n, n]$ over $\mathbb{F}_{2^m}$ cost 131.6, 241.0, 330.8 and 430.4 bits under CryptographicEstimators ($\omega = 2.81$; the best algorithm is the guessing-enhanced GRS enumeration at the three larger sets, MaxMinors at the smallest), against claims of 128/256/384/512: the 256/384/512 sets are 15–82 bits short, slightly more than the 12.7–79 bits reported in kem-03-4 with $\omega = 3$. [Computed].

### 5.4.1 An executed oracle on the BAG-Loong implicit rejection.

The tracker breaks BAG-Piglet through its rejection key (kem-04-2) but files the identical design in BAG-Loong only as a Medium "design concern", reclassified *down* from High on 17 October 2026, with no key recovery attempted. The decapsulation (`loong_kem_decapsulate`) computes, from a received ciphertext $ct = (core, salt)$,

$$m = \mathrm{Dec}(sk, core), \quad g = \mathrm{derive_g}(pk, m, salt), \quad ct' = \mathrm{Enc}(pk, m, g),$$

sets $valid = [\,\text{Dec ok}\,] \wedge [ct' = core]$, and returns $ss = \mathrm{KDF}(valid\,?\,g : \xi,\ ct')$. On the reject branch $ss = \mathrm{KDF}(\xi, ct')$ binds $ct'$, the re-encryption of the *decrypted* message, not the received ciphertext. We compiled the official BAG-Loong-128 reference implementation and, using only its public `kem_keygen`/`kem_enc`/`kem_dec`, measured ([Measured], `oracle_demo.c`): 4000 pairwise-distinct one-bit corruptions of a single valid ciphertext core (salt fixed) all decapsulate to *reject* and return one and the same shared secret, where a sound Fujisaki–Okamoto transform would return 4000 near-independent secrets; the secret is moreover invariant under corruption of up to 256 core bytes but differs for a ciphertext encapsulating a different message. The implicit-rejection secret therefore depends only on the decrypted message, a plaintext-equality / decoding oracle: BAG-Loong's KEM is not IND-CCA2 at the design level, independently of the sampler defect kem-03-3. The same rejection-key route recovers the *BAG-Piglet* key (kem-04-2); we did not carry it through to a key on BAG-Loong, so this result *reinforces* the kem-03-2 oracle rather than demonstrating BAG-Loong key recovery, which remains unshown.

Since the public-support defect (kem-03-3) already recovers BAG-Loong keys from the submitted implementation, and we have now executed the rejection-key oracle on the reference code, the design-level status of BAG-Loong is: a plaintext-checking oracle leading to key recovery (as on BAG-Piglet), and parameters below the claimed level at 256/384/512. *Verdicts: BAG-Piglet previously broken; BAG-Loong newly broken here at the design level (implicit-rejection oracle, executed), also previously broken at the implementation level and below claim (parameters).*

## 6 UVW-KEM and the UVW signature

UVW-KEM (kem-38) encrypts with a masked Reed–Solomon code in a $(U{+}V, U{+}W)$ construction with hidden column pairs ("twins") and relies on the Generalized $(U{+}V, U{+}W)$ Code Distinguishing assumption (Theorem 3 of the specification, claimed exponentially hard when $k_1 = k_2$). The assumption is false: a degree-3, order-2 multiplicity count on shortened public codes of dimension $\Theta(\sqrt{n})$ exhibits an excess of $k_2 - k_l + 2$ cubic forms (800 against 636 generic for UVW-KEM-128 at $k_l = 53$), the dependencies yield the 430 twin pairs and their ratios in three adaptive rounds, and subtracting the twins leaves a $[430, 215]$ GRS code recovered by Sidelnikov–Shestakov; every fresh UVW-KEM-128 ciphertext then decapsulates in 0.4 s from the public key alone (kem-38-8, Critical/Confirmed; distinguishing route by L. Wang, syzygy key recovery and full-size decapsulation by Saarinen; our companion paper [6] develops the structural theory of this cubic-jet attack and its scope). Independently, the implementation's 1000-retry decoder leaves an overall failure rate of $2^{-43.3}/2^{-42.1}/2^{-41.7}$ whose failures occur only on hidden pairs, so that a failure-only reaction recovers all pairs from $\approx 1000$ failures ($\approx 2^{53}$ honest queries; kem-38-5, [4]), a stateful timing reaction recovers UVW-KEM-128 session keys in 2.8 h (kem-38-3), UVW-512's encryption randomness is derived from 256 bits (kem-38-1), and unused ciphertext bits make the submitted code malleable (kem-38-4). *Verdict: previously broken at the design level.*

The UVW signature (sign-32), a Wave-type scheme over $\mathbb{F}_3$, samples its signatures' supports as forced pairs and maps them through the long-term secret monomial transformation; the pairing

persists for the key's lifetime and 300 known-message signatures suffice to recover an equivalent signing key and forge for UVW-128 (Feussner, sign-32-3, Critical/Confirmed; extended to UVW-256 by Saarinen with 308 signatures in 13 minutes). The submitted verifier also accepts every signature (sign-32-1). *Verdict: previously broken.* The forced-pair leak is the signature-side analogue of the twin structure exploited in the KEM; we add nothing to the published attack and record it here only for completeness of the code-based list.

# 7 Conditions for repair

None of the six candidates in the *below claimed level* group is broken: each fails to meet a claimed figure, but a specific, bounded change would return it to *no weakness found*. This section states that change for each, and its cost. The multi-instance repairs are computed with the pinned DS-DOOM estimator (`remediation.py`): for a case that crosses within the $Q = 2^{80}$ same-key budget we report the DS-DOOM message cost *at* $Q = 2^{80}$ and the deficit $\lambda - \mathrm{cost}(2^{80})$, the single-instance margin that must be added so the cost after $2^{80}$ ciphertexts stays at the claimed level. Table 8 collects the conditions.

**The four multi-instance crossings (BIKE-MLThre, QUBE, HARE, TRIKE).** All four cross only because they were parametrised with little headroom above the single-instance target, so the DS-DOOM discount over a static key erases the margin. The deficits at $Q = 2^{80}$ are small: 2.3 bits (BIKE-MLThre-256), 3.5 (HARE-1), 3.7 (BIKE-MLThre-128) and 4.5 (QUBE-128)—a few percent of weight and length. TRIKE is the outlier at 12.4 bits, because TRIKE-1's single-instance $(2, 3)$-QCSD margin is thinnest of all; holding its error density fixed, the smallest repaired length is $r \approx 12075$ (from 10301, a factor 1.17), giving a $[36225, 12075]$ instance whose DS-DOOM cost after $2^{80}$ ciphertexts is $129.98 \geq 128$. BIKE-MLThre additionally must restore the abandoned "2 primitive mod $r$" rule by choosing a different prime of similar size—a structural correction that costs nothing in size—and TRIKE must ship the one-word decoder erratum (min for max, kem-36-1) and fold the boundary weak-key classes into its DFR bound (kem-36-4).

**TriQ-KEM.** Its repair is level-dependent and, at the high levels, not a parameter tweak. Under the standard criterion DFR $< 2^{-\lambda}$, TriQ-128 is short by only 6.1 bits of failure rate, which a modest increase in inner-code redundancy would close; but TriQ-256/384/512 are short by 82.7, 174.4 and 282.0 bits (natural DFR $2^{-173}$ to $2^{-230}$ against a required $2^{-256}$ to $2^{-512}$). Closing a gap of that size means materially stronger error correction—a redesign of the concatenated inner/outer code, not a re-parametrisation. TriQ-KEM at 256 and above is therefore the hardest of the five non-QCTM cases to rescue.

**QCTM.** QCTM is the one case with no clean repair, because its shortfall is a (heuristic) key-recovery estimate rather than an accounting margin. It returns to "standing" only if the 2026 hold-out/GIJS model is shown *not* to apply to the quotient's sparse-support, small-degree regime ($n_b/q \approx 0.002$, $t_0 \in \{15, 27, 53\}$)—a regime no published run has tested—or if $t_0$ and $n_b$ are raised until the quotient key-recovery cost clears the claimed level under the best known model. Neither the growth required nor the attack's true cost in this regime is currently known, which is also why we keep QCTM below claim rather than in the broken row (§4).

**Scope.** The four multi-instance conditions bind only if the same-key multi-ciphertext model is in scope. The NGCC call sets no per-key session cap, so we price to the tracker's $2^{80}$ budget; should

the evaluation cap sessions or exclude that model, BIKE-MLThre, QUBE, HARE and TRIKE are already at "standing" once their decoder, DFR-analysis and $r$-choice errata are applied, and only TriQ-KEM (DFR) and QCTM (structure) remain below claim.

# 8 Conclusions and open problems

Of the fifteen code-based NGCC KEMs, three have had keys or session keys actually recovered—BAG-Piglet by a chosen-ciphertext oracle (kem-04-2), UVW-KEM from the public key alone (kem-38-8), and BAG-Loong from its implementation's public support spaces (kem-03-3)—and a fourth, NSS-HQC, has its 256-, 384- and 512-bit sets 30 to 70 bits below their claimed levels (kem-26-3) with a reported failure-oracle key recovery at 256 (kem-26-2, rated probable); we add that NSS-HQC's *key*-recovery extrapolation is itself non-conservative by 8–19 bits at 256/384 (§3). To BAG-Loong we add a design-level break: we executed, on its reference implementation, the implicit-rejection oracle the tracker had downgraded to a design concern, showing the KEM is not IND-CCA2 regardless of the sampler fix (§5.4.1). One has its security *reduced* by this paper to a single well-defined problem whose status is the subject of the most active debate in code-based cryptography in 2026 (QCTM $\equiv$ Goppa key recovery at $(18, n/19, t/19)$ with subgroup support), seven are below their claimed levels in the same-key multi-ciphertext model that the NGCC call does not exclude (BIKE-MLThre, HARE, NSS-HQC, TriQ-KEM, QUBE, TRIKE, QCTM), one has failure rates that invalidate its IND-CCA2 argument at every level (TriQ-KEM), and four show no design-level weakness under our analysis (BRQC, BRA, C-Multi-UR-AG, Mito), as do HEP-QC-1/3/5 in the precise sense that they are HQC-1/3/5 with a broken but inessential disguise. Every one of the fifteen has at least one finding on the tracker and eleven have confirmed *implementation* defects, several of them total—unseeded generators that make every key public (HEP-QC, BIKE-MLThre, QUBE's optimised build), ciphertext malleability through ignored padding bits (BRA, BRQC, C-Multi-UR-AG, UVW-KEM), a Fujisaki–Okamoto check that compares a prefix (QUBE)—and the gap between "the specification" and "the submitted code" is, for this round, larger than the gap between the specifications and the state of the art.

Three problems are left open and seem worth the community's attention.

1. *Goppa key recovery at small degree with sparse, structured support.* QCTM reduces to binary Goppa codes with $n_b/q \approx 0.002$, $t_0 \in \{15, 27, 53\}$, and support inside a multiplicative subgroup of index 19. All the 2026 structural attacks have been costed and tested for $n \approx q$ and $t \geq 64$, and the one published obstruction [18] weakens as $t$ falls. Three questions follow: do the hold-out and higher-order-vanishing distinguishers see anything at $n_b/q \approx 0.002$; does the subgroup structure of the support help; and what does Apon's bound become at $t_0 = 15$? A positive answer to the first two breaks QCTM outright; a negative one would make QCTM's quotient the first Goppa instance whose hardness rests on a short support rather than on a large degree.
2. *Multi-ciphertext accounting.* Seven of the eight quasi-cyclic candidates set their weights so close to the single-instance target that a static key loses its margin within the $2^{64}$–$2^{80}$ sessions that the call's signature track takes for granted. Whether the evaluation should cap sessions, or the schemes should pay $\approx$ 10–20 bits of weight, is a policy question that the next round should settle explicitly.
3. *When is $\overline{E}$ tight?* This question, which an earlier version of this paper raised as open, is answered by Song, Chen, Zhang, Huang, Weng and Wang [26], who derive the tight bound on the number of independent MM equations for general blocks and its validity conditions (the blocks must be taken in the designers' largest-rank-first order). Our earlier free-ordering, heavy-puncturing search evaluated $\overline{E}$ outside that regime, which is why it appeared to find points below the tab-

ulated minima; those points are artifacts of applying $\overline{E}$ where it is not a tight bound, not real attacks (§5.2). With the correct ordering and the tightness conditions of [26], the designers' published strengths stand. A remaining question of independent interest is the transposed MinRank formulation for $m < n$ (BRA).

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# A Index of forum findings used in this paper

Table 9 lists every tracker finding on the fifteen KEMs and the UVW signature as of 31 October 2026 (identifier, reporter, severity/status, scope: D = design, I = implementation, S = side channel), with a one-line summary. "AI" marks AI-assisted discovery as declared by the reporter.

Table 9: Tracker findings (ngcc.dev), 31 October 2026.

| ID | Reporter | Sev./status | Summary |
|---|---|---|---|
| kem-03-1 | Saarinen (AI) | Med/Conf, S | secret-dependent pivoting in the Gabidulin decoder |
| kem-03-2 | Saarinen (AI) | Med/Conf, D | rejection key does not bind the received ciphertext (plaintext-checking oracle) |
| kem-03-3 | Z. Liu | Crit/Conf, I | reference sampler uses public support spaces; $X$ recovered by linear algebra in $< 1$ s |
| kem-03-4 | Xiong, Wang (AI) | Crit/Conf, D | merged support $t_1{+}t_2{-}g$: $2^{243.3}/2^{333.3}/2^{432.9}$ for 256/384/512 |
| kem-03-5 | Xiong, Wang (AI) | Low/Conf, I | decoder rejects spec-conforming supports |
| kem-04-1 | Saarinen (AI) | Med/Conf, S | secret-dependent pivoting |
| kem-04-2 | Saarinen (AI) | Crit/Conf, D | rejection-key oracle: key recovery with 57,057 (128) / $\approx 3.7 \cdot 10^5$ (256) queries |
| kem-05-1 | Xiong, Wang | Crit/Conf, I | default library's DRBG never seeded: first key pair public |
| kem-05-2 | Saarinen (AI) | Crit/Conf, D | DS-DOOM crossings: $2^{127.89}$ at $2^{69}$ ct (128), $2^{255.78}$ at $2^{73}$ (256) |
| kem-06-1 | Porechna; Xiong, Wang | Low/Conf, I | malformed ciphertexts crash BRA-128/256 |
| kem-06-2 | (ext. of kem-06-1) | Low/Conf, I | field multiplication writes one limb past its output |
| kem-06-3 | Saarinen (AI); Sun (AI) | High/Lead, S | decoder address trace is a rank oracle; $E_y$ recovered with 3.4–6.5k queries (instrumented) |
| kem-06-4 | Saarinen (AI) | Crit/Conf, I | ignored padding bits: 1023 byte-distinct variants per ciphertext |
| kem-06-5 | Saarinen (AI) | Low/Conf, I | key expansion depends on the external DRBG |
| kem-06-6 | LK-PQC-Hunter | Low/Conf, I | short ciphertexts over-read |
| kem-07-1 | Saarinen (AI); Sun (AI) | High/Lead, S | address-trace key-recovery route; modelled $2^{71.4}/2^{86.6}/2^{119.8}$ |
| kem-07-2 | Saarinen (AI) | Crit/Conf, I | ignored padding bits (7/5/1 per vector) |
| kem-07-3 | Saarinen (AI) | Low/Conf, I | key expansion depends on the external DRBG |
| kem-07-4 | LK-PQC-Hunter | Low/Conf, I | short ciphertexts over-read (designers acknowledged) |
| kem-10-1 | Saarinen (AI) | High/Conf, I | malformed ciphertexts crash the decapsulator |
| kem-10-2 | Saarinen (AI) | Low/Conf, S | secret-derived pivots select addresses |
| kem-10-3 | Saarinen (AI) | Crit/Conf, I | ignored padding bits (512 set) |
| kem-10-4 | Saarinen (AI) | Low/Conf, I | key expansion depends on the external DRBG |
| kem-10-5 | LK-PQC-Hunter | Low/Conf, I | short ciphertexts over-read (acknowledged) |
| kem-10-6 | LK-PQC-Hunter | Low/Conf, I | 79-bit field multiplication touches a fourth limb |
| kem-16-1 | Xiong, Wang (AI) | Med/Proof gap, D | Model-2 DFR not reproducible; Model 1 gives $2^{-121.4}$ (128), $2^{-239.2}$ (256) |
| kem-16-2 | Xiong, Wang (AI) | Crit/Conf, D | DS-DOOM crossings at $2^{72}/2^{64}/2^{75}$ ct for 256/384/512 |
| kem-17-1 | Saarinen (AI) | Crit/Conf, I | PRNG never initialised: reproducible secret keys |
| kem-17-2 | Saarinen (AI) | Crit/Conf, D | HEP-QC-7 key pair derived from a 256-bit seed |
| kem-17-3 | Saarinen (AI); Yan et al. | Crit/Conf, I | first encapsulation reproducible; $2^{80}$ state search from two ciphertexts |
| kem-17-4 | T. Xie | High/Conf, D | column multiplicities ($3^{5888}1^{64}$) falsify EPC-P; $T$ and blocks recovered on 35/35 keys |

| ID | Reporter | Sev./status | Summary |
|---|---|---|---|
| kem-17-5 | Saarinen (AI) | Med/Conf, S | secret permutation and mixing matrix drive memory access |
| kem-23-1 | Saarinen (AI) | Med/Conf, I | Mito-E discards erasures: shipped decoder is not the analysed one |
| kem-23-2 | Saarinen (AI) | Low/Conf, I | external DRBG used as XOF |
| kem-26-1 | AMSS | Med/Conf, D | fixed-weight parity falsifies DQCSD/JPCI |
| kem-26-2 | AMSS; Sun (AI) | Crit/Probable, D | DFR analysis invalid (7/50,000 failures); failure-oracle equivalent-key recovery, $\approx 2^{27}$ queries |
| kem-26-3 | Saarinen (AI) | Crit/Conf, D | Stern 234.4/341.5/450.3 raw, 226.5/333.0/441.4 with $\sqrt{n}$, below 256/384/512 |
| kem-26-4 | Xiong, Wang (AI) | Crit/Conf, D | 256-bit decryption sub-seed caps 384/512 at $\approx 2^{280}$ |
| kem-26-5 | Sun (AI) | Crit/Probable, S | RS-decoder timing recovers the NSS-HQC-128 key ($9.1 \cdot 10^5$ queries) |
| kem-32-1 | Saarinen (AI) | Low/Conf, I | trace path retains the secret error vector (fixed by designers) |
| kem-32-2 | Saarinen (AI) | Crit/Conf, D | DS-DOOM crossings $2^{127.66}$ at $2^{79}$ (128), $2^{255.34}$ at $2^{76}$ (256); designers propose $n = 10545/19551$ |
| kem-32-3 | Saarinen (AI) | Crit/Conf, I | bundled DRBG caps QCTM-512 key generation and encapsulation at 384 bits |
| kem-32-4 | L. Wang; Saarinen (AI) | Med/Conf, D | folding exposes an ordinary Goppa quotient $[530, 259]/[1000, 513]/[2000, 1045]$ |
| kem-32-5 | L. Wang | Crit/Lead, D | quotient key recovery $2^{141.3}/2^{142.0}$ (256/512) under Weis's GIJS model; lifting open |
| kem-33-1 | Saarinen (AI) | Med/Conf, S | variable-length sampler re-expands the private seed |
| kem-33-2 | Saarinen (AI) | Med/Conf, S | decryption multiplication traverses the secret support |
| kem-33-3 | Xiong, Wang (AI) | Crit/Conf, I | optimised API never seeds its SHAKE context |
| kem-33-4 | Saarinen (AI) | Crit/Conf, I | optimised FO check compares a prefix only (one-query CCA distinguisher) |
| kem-36-1 | Saarinen (AI); Xiong, Wang | High/Conf, D | specified decoder (max) rejects 1000/1000 honest ciphertexts; code uses min |
| kem-36-2 | Xiong, Wang (AI) | Info/Withdrawn | (binomial notation misread) |
| kem-36-3 | Xiong, Wang (AI) | Low/Conf, I | $h_0$ buffer never freed |
| kem-36-4 | Xiong, Wang (AI) | Med/Proof gap, D | boundary weak-key classes omitted from the DFR bound (95% UCL $2^{-21.7}$) |
| kem-36-5 | Xiong, Wang (AI) | Med/Proof gap, D | Theorem 4 reduction starts from secret data |
| kem-36-6 | Xiong, Wang (AI) | Med/Conf, S | decoder addresses depend on the secret support |
| kem-36-7 | Xiong, Wang (AI) | Info/Conf, D | unsalted FO multi-ciphertext loss $2^l/T$ |
| kem-36-8 | Saarinen (AI) | Low/Conf, I | external DRBG reset for re-encryption |
| kem-37-1 | Saarinen (AI) | Med/Conf, S | variable-work private-support re-expansion |
| kem-37-2 | Xiong, Wang (AI) | Med/Lead, S | FO re-encryption leaks the decrypted message through the bounded-density sampler |
| kem-38-1 | Saarinen (AI) | Crit/Conf, I | UVW-512 encryption pair from 256 bits |
| kem-38-2 | Saarinen (AI) | High/Conf, D | stable list-decoding failure oracle (60× timing gap) |
| kem-38-3 | Xie, Liu (AI) | High/Probable, S | stateful timing reactions recover UVW-128 session keys (2.8 h) |
| kem-38-4 | Saarinen (AI) | Crit/Conf, I | unused $c_1$ bits: ciphertext malleability |
| kem-38-5 | Xiong, Wang (AI) | High/Lead, I | final failures ($2^{-43.3}$) expose the hidden pairs ($\approx 2^{53}$ queries) |
| kem-38-6 | Xiong, Wang (AI) | Low/Conf, I | retry matrices leak memory |
| kem-38-7 | Saarinen (AI) | Low/Conf, I | external DRBG reset |
| kem-38-8 | L. Wang; Saarinen (AI) | Crit/Conf, D | cubic-jet syzygies recover the secret structure; full decapsulation for UVW-128 |

| ID | Reporter | Sev./status | Summary |
|---|---|---|---|
| sign-32-1 | Saarinen (AI) | Crit/Conf, I | every signature accepted |
| sign-32-2 | Saarinen (AI) | Low/Conf, I | all-zero signature crashes the verifier |
| sign-32-3 | Feussner (AI); Saarinen (AI) | Crit/Conf, D | pair-correlated signatures recover an equivalent key (300 signatures) |
| sign-32-4 | Saarinen (AI) | Low/Conf, I | external DRBG reset |

# B Parameter sets

Table 10: Parameter sets as submitted (claimed classical level in parentheses).

| Candidate | Set | Parameters |
|---|---|---|
| HARE | HARE-1 (128) | $n$=16067, $w$=$w_r$=66, $w_e$=105, RS $n_1$=41, RM $n_2$=384, $\alpha$=7 |
| | HARE-2 (128) | $n$=20899, $w$=$w_r$=79, $w_e$=145, $n_1$=32, $n_2$=640, $\alpha$=11 |
| | HARE-3 (192) | $n$=32653, $w$=$w_r$=100, $w_e$=163, $n_1$=51, $n_2$=640, $\alpha$=11 |
| | HARE-5 (256) | $n$=52379, $w$=$w_r$=131, $w_e$=224, $n_1$=81, $n_2$=640, $\alpha$=9 |
| | HARE-7 (384) | $n$=104869, $w$=$w_r$=193, $w_e$=361, $n_1$=91, $n_2$=1152, $\alpha$=15 |
| | HARE-9 (512) | $n$=173981, $w$=$w_r$=259, $w_e$=449, $n_1$=151, $n_2$=1152, $\alpha$=13 |
| HEP-QC | 1/3/5 (128/192/256) | HQC-1/3/5: $n$=17669/35851/57637, $\omega$=66/100/131, $\omega_r$=$\omega_e$=75/114/149, $n_1$=46/56/90, $n_2$=384/640/640 |
| | 7 (512) | $n$=197123, $\omega$=261, $\omega_r$=$\omega_e$=297, $n_1$=220, $n_2$=896, $k$=512 |
| NSS-HQC | 128/256/384/512 | $n$=29443/54493/122579/217901, $w_{sk}$=73/117/175/233, $w_{r_1}$=$w_{r_2}$=66/106/159/213, $w_e$=117/187/279/372, $n_1$=46/70/100/130, $k_1$=16/32/48/64, mult 5 |
| TriQ-KEM | 128/256/384/512 | $n$=16301/50363/97651/157627, $w_x$=$w_y$=$w_r$=67/132/196/260, $w_e$=106/234/350/475, RS [42, 16]/[64, 32]/[108, 48]/[136, 64], RM $n_2$=384/768/896/1152 |
| QUBE | 128 | $r$=13109, $(w_x, w_y)$=(74, 76), $(w_f, w_g, w_e)$=(29, 27, 83), RS [34, 16], RM 384 |
| | 192 | $r$=25357, (110, 113), (43, 39, 121), RS [66, 24], RM 384 |
| | 256 | $r$=41651, (146, 151), (57, 52, 162), RS [108, 32], RM 384 |
| | 384 | $r$=82757, (219, 226), (85, 78, 244), RS [128, 48], RM 640 |
| | 512 | $r$=135851, (291, 301), (113, 103, 324), RS [150, 64], RM 896 |
| Mito | Mito1 128 | $n$=15373 (2-adic), $(w_x, w_y)$=(44, 41; 47, 40), $(w_r, w_t)$=(42, 45; 44, 45), $w_e$=85, RM 384, RS $N_{out}$=40 |
| | Mito1 256 | $n$=33827, (65, 63; 72, 69), (63, 66; 71, 74), $w_e$=133, RM 384, $N_{out}$=88 |
| | Mito1 512 | $n$=106261, (127, 126; 143, 131), (123, 134; 137, 138), $w_e$=262, RM 640, $N_{out}$=166 |
| | Mito1-E 128 | $n$=14621, (44, 41; 47, 40), (42, 45; 44, 45), $w_e$=86, $N_{out}$=38 |
| | Mito1-E 256 | $n$=32261, (69, 64; 72, 64), (63, 73; 66, 72), $w_e$=132, $N_{out}$=84 |
| | Mito1-E 512 | $n$=102437, (128, 123; 140, 136), (126, 129; 136, 141), $w_e$=261, $N_{out}$=160 |
| | Mito2-E 128 | $n$=10253 (4-adic), $w_1$=(25, 25, 22, 21), $w_2$=(22, 23, 25, 25), $w_e$=39, RM 256, $N_{out}$=40 |
| | Mito2-E 256 | $n$=21523, (38, 38, 33, 33), (34, 34, 38, 38), $w_e$=68, RM 256, $N_{out}$=84 |
| | Mito2-E 512 | $n$=65293, (72, 72, 66, 62), (65, 66, 70, 73), $w_e$=135, RM 384, $N_{out}$=170 |
| BIKE-MLThre | 128/256/512 | $r$=12323/40973/150001, $w$=142/274/546, $t$=134/264/524 |
| TRIKE | 2/5/7/9 (128/256/384/512) | $r$=15581/35363/69691/114043, $w$=105/165/249/333, $t$=263/429/659/877; listed but not submitted: TRIKE-1 ($r$=10301, $w$=81, $t$=201), TRIKE-3 ($r$=22003, $w$=123, $t$=319) |

| Candidate | Set | Parameters |
|---|---|---|
| QCTM | 128/256/512 | $m$=18, $\ell$=19, $t_0$=15/27/53, $t$=285/513/1007, $n$=10070/19000/38000, $k$=4957/9783/19891, $w$=142/256/503 |
| BRQC | 128/256/512 | $q$=2, $m$=127/163/229, $n$=119/161/227, $k$=3, $(w_x, w_y, w_{r_1}, w_{r_2}, w_e)$=(5,5,5,5,8)/(5,6,6,7,8)/(6,7,8,8,8), Gabidulin radius $r$=58/79/112 |
| BRA | 128/256/512 | $m$=67/83/127, $n$=121/161/221, $k$=4/4/5, $(w_x, w_y, w_{r_1}, w_{r_2}, w_e)$=(4,5,5,5,8)/(5,5,6,6,8)/(6,6,7,8,8), augmented-Gabidulin radius 53/68/98 |
| C-Multi-UR-AG | 128/256/512 | $m$=79/127/181, $n$=35/45/67, $k$=3/3/4, $N_1$=11/15/19, $N_2$=16/16/21, $w_1$=8/10/12, $w_2$=9/11/14 |
| BAG-Loong | 128/256/384/512 | $n$=42/65/83/104, $m$=47/67/83/97, $n'$=46/67/83/97, $\epsilon$=33/51/66/73, $k$=3/4/5/6, $(t_1, g, t_2)$=(5,4,5)/(5,3,6)/(6,3,6)/(7,4,7) |
| BAG-Piglet | 128/256/384/512 | $n$=47/78/106/125, $m$=43/79/103/131, $\epsilon$=30/54/77/92, $k$=3/4/4/4, $w_{(x,y)}$=(4,4,4)/(6,6,6)/(7,7,7)/(8,8,8), $w_{(r_1,R_2,e)}$=(2,3,3,3)/(3,3,4,4)/(4,4,4,4)/(4,4,5,5) |
| UVW-KEM | 128/256/512 | $q$=433/857/1709, $n$=860/1708/3412, $k_1$+$k_2$=430/854/1706, $w$=116/232/463 |

# C Scripts

Every computation in this paper was carried out by one of the following scripts, available at `https://github.com/yonggewang/ClaudeAI/tree/main/NGCC` and named here so that each claim can be traced to the code that produced it (Python 3; `cryptographic-estimators==2.1.1`, `sympy`, `mpmath`, `numpy`, `scipy`; the pinned `doom.py` is fetched and hash-checked; the one exception is the BAG-Loong oracle, a small C driver linked against the candidate's own reference implementation and using only its public KEM API): `ring_audit.py` (§3.1); `price_hamming.py` (Table 3); `nss_key.py` (§3, NSS-HQC key extrapolation); `loong_break/oracle_demo.c` (§5.4.1, BAG-Loong implicit-rejection oracle, executed on the reference implementation); `multi_instance.py` (Table 4); `remediation.py` (§7, Table 8); `dfr_model1.py` (Table 5); `qctm_toy.py` (Theorems 4–7, Table 6); `mm_blockwise.py`, `mm_search.py`, `mm_search_descending.py` (§5.2); `rank_pricing.py` (Table 7); `mito_toy.py` (Proposition 1); `square_toy.py` (§4); `rsl_toy.py` (§5).

# D Design teams

For reference, Table 11 lists the submitters of each candidate as stated on the title page of its Round-1 specification. Four candidates (BAG-Loong, BAG-Piglet, QCTM, QUBE) carry no author block in the public specification—their submitter details sit in the separate "Basic Algorithm Information" form, which is not part of the archived package—so we do not name them. Three rank-metric KEMs (BRA, BRQC, C-Multi-UR-AG) share a single team, and the two UVW schemes share theirs.

Table 11: Submitters as listed in the Round-1 specifications.

| Candidate | Submitters (as listed; * = corresponding / contact) |
|---|---|
| BAG-Loong | not stated in the public specification |
| BAG-Piglet | not stated in the public specification |
| BRA, BRQC, C-Multi-UR-AG | Xinyi Huang, Yongcheng Song*, Rongmao Chen, Jiang Zhang, Debiao He, Xiaofeng Chen, Junzuo Lai, Yi Wang, Chao Lin, Jiang Tan, Kang Lin (one team; affiliations not listed in the specs) |

| Candidate | Submitters |
|---|---|
| HARE | Tianrui Wang (Tsinghua, Inst. for Network Sciences and Cyberspace), Qicheng Teng (Tsinghua IAS), Jipeng Li (Xidian), Anyu Wang (Tsinghua IAS), Jun Zhang (Capital Normal), Bo Pang, Chaojian Shen (ICBC) |
| HEP-QC | Chengju Li*, Tianrui Liu, Chunyu Gan, Zhongyu Huang, Shuying Dong, Xinran Wang (East China Normal University) |
| Mito | Changlun Li, Chengju Li, Liang Bai, Qiping Lin, Tianrui Liu, Yan Ding, Chunyu Gan, Zhongyu Huang (Beijing Key Lab of Post-Quantum Cryptography; Xingtang Telecom; East China Normal University) |
| NSS-HQC | Yulong Shen, Xutong Mu, Shuaiping Dong, Zidong Zhou, Ahmat Bahadir, Junbo Ni, Haoyu Zhang, Jinxuan Cui, Ke Cheng, Anxiao Song, Zhenxing Niu, Lele Zheng (Xidian University) |
| QCTM | not stated in the public specification |
| QUBE | not stated in the public specification |
| TRIKE | Tianrui Wang (Tsinghua INSC), Yuxuan Li (SUSTech), Jing Tian (Nanjing Univ.), Yangcan Zhou (Sun Yat-sen Univ.), Zijian Zhang (Nanjing Univ.), Anyu Wang* (Tsinghua IAS) |
| TriQ-KEM | Lan Luo*, Chaofeng Guan, Yunlei Zhao, Hong Wang, Jianhua Hou, Tong Yu |
| UVW-KEM | Fangguo Zhang, Yang Yang, Kuojie Xiao, Xiaofeng Chen, Chen Li, Haibo Tian (Sun Yat-sen Univ.; Jiangxi Normal Univ.; Xidian Univ.) |
| UVW signature | Fangguo Zhang, Yang Yang, Yixuan Huang, Xiaofeng Chen, Chen Li, Haibo Tian (same affiliations) |

Two cross-team ties are worth noting: Tianrui Wang and Anyu Wang lead both HARE and TRIKE, and Xiaofeng Chen appears on both the BRA/BRQC/C-Multi-UR-AG team and the UVW team.

Table 2: Verdicts. *PB*: previously broken (forum); *NEW*: new reduction or shortfall in this paper; *BC*: below the claimed level (parameters); *NW*: no design-level weakness found. I = implementation-level only. Details and labels in the sections cited.

| Candidate | Verdict | Basis | § |
|---|---|---|---|
| BAG-Loong | PB, BC | key recovered from public supports in $< 1$s (kem-03-3); implicit-rejection oracle also executed here (kem-03-2) [Measured]; merged support 241/331/430 $<$ 256/384/512 [Computed] | 5 |
| BAG-Piglet | PB | CCA key recovery, $5.7 \cdot 10^4$ queries (kem-04-2) | 5 |
| BIKE-MLThre | BC, NEW | multi-instance 127.9/255.8 (kem-05-2, confirmed); $r = 150001$ not 2-primitive [Computed] | 3 |
| BRA | NW (I) | blockwise MM 189/304/556 reproduced in the tight-$\overline{E}$ regime ([26]) [Computed]; padding malleability (kem-06-4) | 5 |
| BRQC | NW (I) | blockwise MM 181/295/566 reproduced in the tight-$\overline{E}$ regime ([26]) [Computed]; padding malleability (kem-07-2) | 5 |
| C-Multi-URAG | NW (I) | RSD 197/313/571 vs claimed 170/276/530 ($\omega$) [Computed]; RSL stacking saturates [Measured] | 5 |
| HARE | BC, NEW | DFR proof gap (kem-16-1); multi-instance 256/384/512 (kem-16-2) and HARE-1 at $2^{66}$ [Computed] | 3 |
| HEP-QC | PB, BC | EPC-P falsified (kem-17-4); HEP-QC-7 capped at $2^{256}$ (kem-17-2); otherwise =HQC | 3 |
| Mito | NW (I) | margins $\geq 65$ bits; 2-adic projection costs 2 bits [Computed]; Mito-E decoder not implemented (kem-23-1) | 3 |
| NSS-HQC | PB, NEW | ISD 226/333/441 (kem-26-3); key extrapolation $1.0969\sqrt{n}$ non-conservative, 248/364 vs 256/384 [Computed]; DFR invalid (kem-26-2) | 3 |
| QCTM | BC, NEW | fold = Goppa $(18, n/19, t/19)$ and poly-time lifting [Proved][Measured]; quotient estimates $2^{141}$ (kem-32-5); multi-instance (kem-32-2) | 4 |
| QUBE | BC, NEW | multi-instance crossings at $2^{71}/2^{72}/2^{74}/2^{80}$ [Computed]; DOOM ignored by the designers | 3 |
| TRIKE | BC, NEW | TRIKE-1 crosses at $2^{31}$ ciphertexts, others $2^{46}$–$2^{63}$ [Computed]; decoder/DFR gaps (kem-36-1/4) | 3 |
| TriQ-KEM | BC, NEW | same-key crossings (kem-37-3) [Computed]; natural DFR $2^{-121.9} \ldots 2^{-230.0}$ meets the spec's criterion but misses a generic $q/2^{\lambda}$ term [Cited] | 3 |
| UVW-KEM | PB | cubic-jet key recovery (kem-38-8, companion paper); DFR reaction (kem-38-5) | 6 |
| UVW signature | PB | pair-correlated signatures recover the key (sign-32-3) | 6 |

Table 3: Single-instance decoding costs ($\log_2$ bit operations, CryptographicEstimators 2.1.1): raw Stern cost; minimum over Prange, Stern, Dumer, ball-collision, BJMM, BJMM$^+$ and May–Ozerov, with the algorithm attaining it; the quasi-cyclic discount and the discounted minimum. The last column is the value the specification publishes for the same quantity, where it publishes one. Instances: *key* = codeword finding, *msg*/*ct* = syndrome decoding, *pk* = the $[2n, n]$ key instance, *ct u-only* = the ciphertext instance that discards $v$. The table covers the instances for which a specification publishes a comparable number, together with the lowest-level instance of each family, up to a size cap; for the larger instances we quote the specifications' own values, which the agreement below licenses.

| Scheme | Set | Claim | Instance | $n$ | $k$ | $w$ | Stern | min (alg.) | disc. | min−disc. | Spec. |
|---|---|---|---|---|---|---|---|---|---|---|---|
| QCTM | 128 | 128 | msg SD [n,k] | 10070 | 4957 | 142 | 154.64 | 152.99 (MO) | 2.12 | 150.87 | 150.87 T1–3 |
| | 256 | 256 | msg SD [n,k] | 19000 | 9783 | 256 | 282.76 | 275.03 (MO) | 2.12 | 272.91 | 272.90 T1–3 |
| HQC | 128 | 143 | ct u-only | 35338 | 17669 | 150 | 169.74 | 169.74 (St) | 7.05 | 162.69 | — |
| | 128 | 143 | pk 2-QCSD | 35338 | 17669 | 132 | 152.39 | 152.39 (St) | 7.05 | 145.34 | — |
| HEP-QC | 128 | 128 | ct u-only | 35338 | 17669 | 150 | 169.74 | 169.74 (St) | 7.05 | 162.69 | — |
| | 128 | 128 | pk 2-QCSD | 35338 | 17669 | 132 | 152.39 | 152.39 (St) | 7.05 | 145.34 | — |
| HARE | 128 | 128 | ct u-only | 32134 | 16067 | 132 | 152.02 | 152.02 (St) | 6.99 | 145.03 | — |
| | 128 | 128 | ct u-only | 41798 | 20899 | 158 | 178.13 | 178.13 (St) | 7.18 | 170.95 | — |
| | 128 | 128 | pk 2-QCSD | 32134 | 16067 | 132 | 152.02 | 152.02 (St) | 6.99 | 145.03 | — |
| | 128 | 128 | pk 2-QCSD | 41798 | 20899 | 158 | 178.13 | 178.13 (St) | 7.18 | 170.95 | — |
| NSS-HQC | 128 | 128 | ct u-only | 58886 | 29443 | 132 | 154.43 | 154.43 (St) | 7.42 | 147.01 | — |
| | 128 | 128 | pk 2-QCSD | 58886 | 29443 | 146 | 167.89 | 167.89 (St) | 7.42 | 160.47 | 188.2 S4.5 |
| TriQ-KEM | 128 | 128 | ct u-only | 32602 | 16301 | 134 | 154.00 | 154.00 (St) | 7.00 | 147.00 | — |
| | 128 | 128 | pk 2-QCSD | 32602 | 16301 | 134 | 154.00 | 154.00 (St) | 7.00 | 147.00 | — |
| QUBE | 128 | 128 | ct u-only (3,2) | 39327 | 26218 | 85 | 154.43 | 154.43 (St) | 6.84 | 147.59 | — |
| | 128 | 128 | pk (3,1)-QCSD | 39327 | 13109 | 224 | 152.61 | 152.61 (St) | 6.84 | 145.77 | 144.77 T6 |
| Mito1 | 128 | 128 | pk full [4n,2n] | 61492 | 30746 | 172 | 193.20 | 193.20 (St) | 7.45 | 185.75 | — |
| | 128 | 128 | pk projected [2n,n] | 30746 | 15373 | 172 | 190.57 | 189.91 (MO) | 6.95 | 182.96 | — |
| Mito1-E | 128 | 128 | ct u full [4n,2n] | 58484 | 29242 | 176 | 196.88 | 196.88 (St) | 7.42 | 189.46 | — |
| | 128 | 128 | pk full [4n,2n] | 58484 | 29242 | 172 | 193.00 | 193.00 (St) | 7.42 | 185.58 | — |
| | 128 | 128 | pk projected [2n,n] | 29242 | 14621 | 172 | 190.39 | 189.57 (MO) | 6.92 | 182.65 | — |
| Mito2-E | 128 | 128 | pk projected [2n,n] | 20506 | 10253 | 186 | 202.86 | 200.81 (MO) | 6.66 | 194.15 | — |
| BIKE-MLThre | 128 | 128 | key 2-QCCF [2r,r] | 24646 | 12323 | 142 | 160.65 | 160.39 (MO) | 13.59 | 146.80 | — |
| | 128 | 128 | msg 2-QCSD [2r,r] | 24646 | 12323 | 134 | 152.92 | 152.92 (St) | 6.79 | 146.13 | — |
| TRIKE | 128 | 128 | key 3-QCCF [3r,2r] | 30903 | 20602 | 81 | 147.38 | 147.38 (St) | 13.33 | 134.05 | 134.05 T9 |
| | 128 | 128 | key 3-QCCF [3r,2r] | 46743 | 31162 | 105 | 185.71 | 185.71 (St) | 13.93 | 171.78 | 171.78 T9 |
| | 128 | 128 | msg (2,3)-QCSD [3r,r] | 30903 | 10301 | 201 | 138.78 | 138.78 (St) | 6.67 | 132.11 | 132.11 T10 |
| | 192 | 192 | key 3-QCCF [3r,2r] | 66009 | 44006 | 123 | 214.79 | 214.79 (St) | 14.43 | 200.36 | 200.36 T9 |

Table 4: Same-key multi-instance message recovery (DS-DOOM, pinned estimator). Instance $[n, k]$, weight $w$, rotations per ciphertext; single-instance cost ($M$ =rot); first session exponent $s$ with $M = \mathrm{rot} \cdot 2^s$ targets below the claimed level, with the time and $\log_2$ memory there; and the cost after $2^{64}$ sessions. Boldface marks a crossing within $2^{80}$.

| Scheme | Set | Target | $n$ | $w$ | rot | Single | $s$ | Time (Mem) | at $2^{64}$ |
|---|---|---|---|---|---|---|---|---|---|
| HARE | HARE-1 | 128 | 32134 | 132 | 16067 | 154.4 | $2^{66}$ | **127.8** (94) | 128.5 |
| | HARE-2 | 128 | 41798 | 158 | 20899 | 180.8 | — | — | 154.8 |
| | HARE-3 | 192 | 65306 | 200 | 32653 | 223.6 | $2^{78}$ | **191.6** (108) | 197.1 |
| | HARE-5 | 256 | 104758 | 262 | 52379 | 286.5 | $2^{72}$ | **255.4** (103) | 259.1 |
| | HARE-7 | 384 | 209738 | 386 | 104869 | 411.4 | $2^{64}$ | **383.2** (97) | 383.2 |
| | HARE-9 | 512 | 347962 | 518 | 173981 | 544.5 | $2^{75}$ | **511.7** (110) | 516.2 |
| HEP-QC | HEP-QC-1 | 128 | 35338 | 150 | 17669 | 172.2 | — | — | 146.3 |
| | HEP-QC-3 | 192 | 71702 | 228 | 35851 | 251.8 | — | — | 224.7 |
| | HEP-QC-5 | 256 | 115274 | 298 | 57637 | 322.7 | — | — | 294.3 |
| | HEP-QC-7 | 512 | 394246 | 594 | 197123 | 620.8 | — | — | 591.6 |
| NSS-HQC | NSS-HQC-128 | 128 | 58886 | 132 | 29443 | 156.0 | $2^{77}$ | **127.6** (107) | 132.1 |
| | NSS-HQC-256 | 256 | 108986 | 212 | 54493 | 237.1 | $2^{0}$ | **237.1** (33) | 211.0 |
| | NSS-HQC-384 | 384 | 245158 | 318 | 122579 | 344.6 | $2^{0}$ | **344.6** (35) | 318.2 |
| | NSS-HQC-512 | 512 | 435802 | 426 | 217901 | 454.0 | $2^{0}$ | **454.0** (37) | 427.2 |
| TriQ-KEM | TriQ-128 | 128 | 32602 | 134 | 16301 | 156.5 | $2^{73}$ | **127.5** (101) | 130.5 |
| | TriQ-256 | 256 | 100726 | 264 | 50363 | 288.3 | $2^{77}$ | **255.6** (108) | 260.8 |
| | TriQ-384 | 384 | 195302 | 392 | 97651 | 417.8 | $2^{77}$ | **383.4** (110) | 389.2 |
| | TriQ-512 | 512 | 315254 | 520 | 157627 | 546.9 | $2^{78}$ | **511.5** (113) | 517.2 |
| QUBE | QUBE-128 | 128 | 39327 | 85 | 13109 | 157.4 | $2^{71}$ | **127.8** (98) | 131.0 |
| | QUBE-192 | 192 | 76071 | 125 | 25357 | 222.4 | $2^{72}$ | **191.4** (101) | 194.7 |
| | QUBE-256 | 256 | 124953 | 166 | 41651 | 287.5 | $2^{74}$ | **255.9** (105) | 259.7 |
| | QUBE-384 | 384 | 248271 | 248 | 82757 | 419.1 | $2^{80}$ | **383.5** (113) | 391.0 |
| | QUBE-512 | 512 | 407553 | 329 | 135851 | 548.5 | — | — | 519.5 |
| Mito1 | Mito1-128 | 128 | 61492 | 176 | 30746 | 199.6 | — | — | 174.1 |
| | Mito1-256 | 256 | 135308 | 274 | 67654 | 298.8 | — | — | 272.1 |
| | Mito1-512 | 512 | 425044 | 532 | 212522 | 559.4 | — | — | 531.2 |
| Mito1-E | Mito1-E-128 | 128 | 58484 | 176 | 29242 | 199.4 | — | — | 173.9 |
| | Mito1-E-256 | 256 | 129044 | 274 | 64522 | 299.2 | — | — | 271.7 |
| | Mito1-E-512 | 512 | 409748 | 532 | 204874 | 559.2 | — | — | 530.8 |
| Mito2-E | Mito2-E-128 | 128 | 82024 | 190 | 41012 | 214.8 | — | — | 188.4 |
| | Mito2-E-256 | 256 | 172184 | 288 | 86092 | 313.9 | — | — | 286.4 |
| | Mito2-E-512 | 512 | 522344 | 548 | 261172 | 576.2 | — | — | 548.2 |
| BIKE-MLThre | BIKE-MLThre-128 | 128 | 24646 | 134 | 12323 | 155.5 | $2^{69}$ | **127.9** (96) | 129.4 |
| | BIKE-MLThre-256 | 256 | 81946 | 264 | 40973 | 288.2 | $2^{73}$ | **255.8** (104) | 260.0 |
| | BIKE-MLThre-512 | 512 | 300002 | 524 | 150001 | 550.6 | — | — | 521.6 |
| TRIKE | TRIKE-1 (not submitted) | 128 | 30903 | 201 | 10301 | 140.9 | $2^{31}$ | **127.9** (59) | 116.2 |
| | TRIKE-2 | 128 | 46743 | 263 | 15581 | 177.2 | — | — | 151.5 |
| | TRIKE-3 (not submitted) | 192 | 66009 | 319 | 22003 | 210.6 | $2^{46}$ | **191.5** (76) | 184.3 |
| | TRIKE-5 | 256 | 106089 | 429 | 35363 | 276.1 | $2^{47}$ | **255.7** (78) | 249.3 |
| | TRIKE-7 | 384 | 209073 | 659 | 69691 | 411.5 | $2^{63}$ | **384.0** (96) | 383.4 |
| | TRIKE-9 | 512 | 342129 | 877 | 114043 | 540.0 | $2^{63}$ | **511.4** (98) | 511.4 |
| QCTM | QCTM128 | 128 | 10070 | 142 | 19 | 162.3 | $2^{79}$ | **127.6** (96) | 133.4 |
| | QCTM256 | 256 | 19000 | 256 | 19 | 291.4 | $2^{76}$ | **255.3** (96) | 260.7 |
| | QCTM512 | 512 | 38000 | 503 | 19 | 562.6 | — | — | 531.9 |
| HQC(ref) | HQC-128 | 143 | 35338 | 150 | 17669 | 172.2 | $2^{74}$ | **142.4** (102) | 146.3 |
| | HQC-192 | 207 | 71702 | 228 | 35851 | 251.8 | — | — | 224.7 |
| | HQC-256 | 272 | 115274 | 298 | 57637 | 322.7 | — | — | 294.3 |

Table 5: Decryption-failure rates ($-\log_2$). "Spec" is the value claimed in the specification (with its model); "Model 1" is our recomputation (`dfr_model1.py`); HARE's Model-1 values are from kem-16-1.

| Scheme | Set | Spec | Model 1 | Note |
|---|---|---|---|---|
| HQC (control) | 1/3/5 | 132.89 / 193.86 / 260.59 | 132.892 / 193.861 / 260.597 | reproduces the QUBE/HQC tables |
| QUBE | 128/192/256/384/512 | 137.5 / 200.1 / 264.3 / 392.4 / 520.6 | 137.545 / 200.136 / 264.313 / 392.436 / 520.668 | reproduced |
| TriQ-KEM | 128/256/384/512 | 121.90 / 173.35 / 209.57 / 230.00 (Table 6, natural) | — (not in our Model-1 script) | spec's own criterion met; short of generic $q/2^\lambda$ |
| HEP-QC | 1/3/5 | $<$ 128/192/256 | 132.892 / 193.861 / 260.597 | =HQC |
| HEP-QC | 7 | $<$ 512 | 574.372 | |
| Mito1 | 128/256/512 | 173.34 / 272.79 / 516.22 | 173.447 / 272.508 / 515.614 | from the spec's $p^*$; $p^*$ reproduced |
| HARE | 1/2/3/5 | $<$ 128/130/194/258 (Model 2) | 121.45 (HARE-2), 239.21 (HARE-5) [kem-16-1] | Model-2 artifact missing |
| HARE | 7/9 | $<$ 384/519 (Model 1) | 384.93 / 519.45 [kem-16-1] | reproduced by kem-16-1 |
| NSS-HQC | 256 | $<$ 128 | meas. $1.4 \cdot 10^{-4}$ [kem-26-2] | analysis invalid |
| Mito1-E / Mito2-E | all | 167.99–547.37 (erasure) | — | decoder not implemented (kem-23-1) |
| TRIKE | 2/5/7/9 | 128/256/384/512 (sim.) | — | boundary classes omitted (kem-36-4) |
| BIKE-MLThre | 128/256/512 | $\leq$ 128/256/512 (target) | — | iteration-wise statistics only |
| UVW-KEM | 128/256/512 | negligible | 43.3 / 42.1 / 41.7 [kem-38-5] | 1000-retry decoder |

Table 6: Lifting experiments (`qctm_toy.py`; seeds 1–3 for the first row).

| $m$ | $\ell$ | $t_0$ | $n_b$ | $n$ | $w$ | $u$ | $\dim \varphi(\mathcal{C})$ | $\dim K_1 = \sum_k (d_k{+}1)$ | lifted & decoded |
|---|---|---|---|---|---|---|---|---|---|
| 10 | 11 | 3 | 48 | 528 | 16 | 321 | 17 | 29 | yes (3/3 seeds) |
| 10 | 3 | 5 | 100 | 300 | 7 | 141 | 49 | 45 | yes |
| 10 | 31 | 3 | 33 | 1023 | 46 | 921 | 2 | 30 | yes |
| 11 | 23 | 3 | 40 | 920 | 34 | 749 | 6 | 33 | yes |
| 12 | 13 | 3 | 80 | 1040 | 19 | 457 | 43 | 35 | yes |

Table 7: Rank-decoding costs under CryptographicEstimators 2.1.1 (`RankSDEstimator`, $\omega = 2.81$, $\log_2$ bit operations): best algorithm, and the MaxMinors cost with its hybrid/puncturing parameters. C-Multi-UR-AG instances are the specification's RSD instances; BAG-Loong uses the merged rank $t_1 + t_2 - g$ (kem-03-4).

| Scheme | Instance | RSD$(q, m, n, k, r)$ | Best (algorithm) | MM | Claim |
|---|---|---|---|---|---|
| C-Multi-UR-AG-128 | key ($w_1$) | $(2, 79, 70, 35, 8)$ | 197.5 (MM, $a$=14) | 197.5 | 170 ($\omega$=2: 172) |
| | ciphertext ($w_2$) | $(2, 79, 81, 35, 9)$ | 200.0 (MM, $a$=11) | 200.0 | 170 |
| C-Multi-UR-AG-256 | key | $(2, 127, 90, 45, 10)$ | 327.1 (MM, $a$=22) | 327.1 | 276 (296) |
| | ciphertext | $(2, 127, 105, 45, 11)$ | 312.6 (MM, $a$=17, $p$=1) | 312.6 | 276 |
| C-Multi-UR-AG-512 | key | $(2, 181, 134, 67, 12)$ | 570.7 (MM, $a$=36, $p$=1) | 570.7 | 530 |
| | ciphertext | $(2, 181, 153, 67, 14)$ | 628.4 (MM, $a$=33) | 628.4 | 530 (580) |
| BAG-Loong-128 | key, $r$=6 | $(2, 47, 84, 42, 6)$ | 131.6 (MM, $a$=9) | 131.6 | 128 |
| BAG-Loong-256 | key, $r$=8 | $(2, 67, 130, 65, 8)$ | 241.0 (GE-GRS) | 306.5 | 256 |
| BAG-Loong-384 | key, $r$=9 | $(2, 83, 166, 83, 9)$ | 330.8 (GE-GRS) | 439.9 | 384 |
| BAG-Loong-512 | key, $r$=10 | $(2, 97, 208, 104, 10)$ | 430.4 (GE-GRS) | 623.4 | 512 |

Table 8: What each "below claimed level" candidate needs to reach "standing". Deficits are DS-DOOM message cost at $Q = 2^{80}$ below the claimed level (`remediation.py`); DFR deficits are against the standard IND-CCA2 criterion DFR $< 2^{-\lambda}$.

| Candidate | Below claim because | Condition to reach "standing" | Cost of the fix |
|---|---|---|---|
| BIKE-MLThre | multi-ciphertext crossing; $x^r - 1$ reducible (40 factors) | +2.3–3.7 bits of margin; pick $r$ with 2 primitive (the design rule) | a few %; new prime $r$ (no size cost) |
| QUBE | multi-ciphertext crossing (DOOM unaccounted) | +4.5 bits of margin; account for DOOM | a few % |
| HARE | multi-ciphertext crossing; Model-2 DFR not reproducible | +3.5 bits of margin; a reproducible DFR analysis | a few % + re-analysis |
| TRIKE | message crossing at $2^{31}$ ct; decoder spec bug (kem-36-1) | +12.4 bits of margin ($r$≈12075, ×1.17); decoder erratum; boundary DFR classes | ∼17% larger keys/ct + errata |
| TriQ-KEM | natural DFR $> 2^{-\lambda}$ at every level | DFR $< 2^{-\lambda}$: short 6.1 bits at 128 but 82.7/174.4/282.0 at 256/384/512 | small at 128; redesign at 256+ |
| QCTM | quotient Goppa key recovery $< \lambda$ under the 2026 hold-out/GIJS model (heuristic, via the lifting theorem) | model shown inapplicable in the sparse-support regime ($n_b/q$≈0.002), *or* $t_0, n_b$ raised until the quotient cost exceeds $\lambda$ under the best model | structural; no clean parameter bump, growth unknown |